

Equations of State for Vehicular Trajectory Completion in Bounded Phase Space

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Abstract

We derive, from the single axiom that a vehicle’s phase space is bounded ($V_{\Gamma} < \infty$), the complete equations of state governing vehicular trajectories. The bounded phase space of an automobile—enforced by physical speed limits, acceleration constraints, and the finite extent of the road network—guarantees Poincaré recurrence, which in turn generates oscillatory dynamics admitting a categorical description. We construct the partition coordinates (n, ℓ, m, s) for vehicular dynamics, where n encodes road hierarchy level, ℓ the directional state, m the lateral displacement, and s the traffic chirality. We prove that the vehicle state capacity at level n satisfies $C(n) = 2n^2$. The S-entropy coordinates $\mathbf{S} = (S_k, S_t, S_e) \in [0, 1]^3$, encoding knowledge, temporal, and evolution entropy respectively, provide a complete categorical characterization of the driving state. Our central result is the *vehicular equation of state* $P_{\text{drive}} \cdot V_{\text{road}} = N \cdot k_B \cdot T_{\text{cat}} \cdot \mathcal{S}$, where P_{drive} is the driving pressure, V_{road} the accessible maneuvering volume, T_{cat} the categorical temperature measuring the rate of partition traversal, and $\mathcal{S}$ a structural factor encoding the partition coordinate distribution. We demonstrate that this equation of state reproduces known traffic flow relations (Greenshields, Lighthill–Whitham) as limiting cases, predicts congestion as a genuine phase transition, and provides a complete trajectory description with complexity $O(\log_3 N)$ rather than the $O(N)$ required by forward simulation. The Position-Partition Bijection $\Pi_V : \text{Road} \times \mathbb{R}^3 \rightarrow [0, 1]^3$ is constructed explicitly and shown to be almost-everywhere invertible. This is the second paper (hard physics) in a three-paper series establishing vehicular trajectory completion from bounded phase space.

1 Introduction

The problem of autonomous driving has been formulated, almost universally, as one of *prediction*: given the current

state of the vehicle and its environment, simulate forward in time to determine the future trajectory [1, 2, 3]. This formulation is natural from the standpoint of Newtonian mechanics, where the equations of motion are integrated forward from initial conditions. However, it suffers from two fundamental pathologies that no amount of engineering refinement can cure.

The first pathology is computational. A vehicle operating in an environment with n interacting agents over a time horizon t at resolution Δt requires $O(n^2 t / \Delta t)$ operations for trajectory prediction [4]. For realistic urban driving scenarios with $n \sim 10^2$ agents, time horizons $t \sim 10$ s, and resolutions $\Delta t \sim 0.01$ s, this yields $\sim 10^7$ operations per planning cycle. While tractable for single-agent planning, the combinatorial explosion in multi-agent interaction renders true prediction intractable at the timescales required for safe driving.

The second pathology is physical. Vehicular trajectories in complex traffic are governed by dynamics that exhibit sensitive dependence on initial conditions. Small perturbations ε_0 in the initial state grow as $\varepsilon(t) \sim \varepsilon_0 e^{\lambda t}$, where $\lambda > 0$ is the maximal Lyapunov exponent of the traffic dynamics [5]. For typical traffic flows, $\lambda \sim 0.5 \text{ s}^{-1}$ [4], so that prediction uncertainty doubles every ~ 1.4 s. Beyond a Lyapunov horizon $t_L \sim \lambda^{-1} \ln(\varepsilon_{\text{tol}}/\varepsilon_0) \sim 5$ s, no forward prediction can be reliable regardless of computational resources. The entire enterprise of prediction-based autonomous driving is thus fundamentally limited by chaos.

We propose a radical alternative. The vehicle’s trajectory is not to be *predicted* but *completed*—that is, determined by the equations of state that govern the system’s evolution in phase space. This is not a semantic distinction. Prediction requires integration of dynamics forward from initial conditions, a process that is both computationally expensive and physically unstable. Completion, by contrast, exploits the global structure of the bounded phase space to determine the trajectory from categorical considerations alone.

The key insight, developed across a series of 23 prior papers [6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19], is that a single axiom—the boundedness of phase space, $V_\Gamma < \infty$ —suffices to derive the complete physical description. Bounded phase space guarantees Poincaré recurrence [20], which generates oscillatory dynamics. Oscillatory dynamics admit categorical description through partition coordinates. The Triple Equivalence

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_B M \ln n \quad (1)$$

identifies the oscillatory, categorical, and partition descriptions as three views of a single entropy [11]. Here M is the partition depth and n the capacity parameter.

The present paper is the second in a three-paper series on vehicular trajectory completion. The first paper [6] establishes the general framework of trajectory completion computing and proves the fundamental identity $O(x) \equiv C(x) \equiv P(x)$: observation, computing, and physical processing are categorically identical operations. The present paper derives, from first principles, the *equations of state* governing a vehicle’s trajectory. The third paper addresses practical implementation.

Our contributions are as follows:

1. We prove that the vehicular phase space Γ_V is bounded and compute the finite state count $N_{\text{max}} = V_\Gamma/h^3$ explicitly (Section 2).
2. We construct the partition coordinates (n, ℓ, m, s) for vehicular dynamics and prove the state capacity $C(n) = 2n^2$ (Section 3).
3. We define the S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ and prove S-completeness (Section 4).
4. We derive the vehicular equation of state $P_{\text{drive}} \cdot V_{\text{road}} = N \cdot k_B \cdot T_{\text{cat}} \cdot \mathcal{S}$ and demonstrate that it reproduces known traffic flow relations (Section 5).
5. We derive the S-entropy evolution equations and prove bounded divergence, zero categorical Lyapunov exponent, and stability (Section 6).
6. We construct the Position-Partition Bijection Π_V and prove almost-everywhere bijectivity (Section 7).
7. We classify the completion morphisms for driving and prove the selection rules (Section 8).
8. We derive multi-system coupling equations for vehicle–vehicle and vehicle–environment interactions (Section 9).
9. We make five quantitative predictions testable with existing traffic data (Section 10).

Throughout, every major claim is stated as a formal theorem with complete proof. We make no approximations, no heuristic arguments, and no appeals to simulation. The equations of state follow from the bounded phase space axiom alone.

2 Bounded Phase Space of an Automobile

2.1 The Fundamental Axiom

We begin with the single axiom upon which the entire theory rests.

Axiom 2.1 (Bounded Vehicular Phase Space). *The phase space of any automobile operating on a physical road network has finite Liouville volume:*

$$V_\Gamma = \int_{\Gamma_V} \prod_{i=1}^f dq_i dp_i < \infty, \quad (2)$$

where f is the number of degrees of freedom and Γ_V is the accessible phase space.

This axiom is not assumed—it is *physically enforced* by three independent constraints:

1. **Spatial boundedness:** The vehicle position q is confined to the road network $\mathcal{R} \subset \mathbb{R}^3$, which has finite total length L_{tot} , finite width w , and finite extent. Thus $\int_{\mathcal{R}} d^3q < \infty$.
2. **Velocity boundedness:** The vehicle speed is bounded by $|v| \leq v_{\text{max}}$, where v_{max} is determined by engine power, aerodynamic drag, and legal limits. Thus $|p| \leq mv_{\text{max}} < \infty$.
3. **Acceleration boundedness:** The vehicle acceleration satisfies $|a| \leq a_{\text{max}}$, where a_{max} is determined by tire-road friction, engine torque, and braking capacity. This bounds the rate of momentum change: $|\dot{p}| \leq ma_{\text{max}} < \infty$.

Definition 2.2 (Vehicular Phase Space). *The vehicular phase space is the set*

$$\Gamma_V = \{(q, p) : q \in \mathcal{R}, |p| \leq mv_{\text{max}}\}, \quad (3)$$

where $\mathcal{R} \subset \mathbb{R}^3$ is the road network manifold, m is the vehicle mass, and v_{max} is the maximum attainable speed.

Remark 2.3. *The road network $\mathcal{R}$ is a one-dimensional manifold (with crossings) embedded in $\mathbb{R}^3$. However, the vehicle has finite extent in the lateral and vertical directions, so the effective configuration space has three position coordinates (longitudinal, lateral, vertical) plus three momentum coordinates. For the purpose of Liouville volume computation, we work in the full six-dimensional phase space.*

2.2 Liouville Volume Computation

Theorem 2.4 (Finite Phase Space Volume). *The Liouville volume of the vehicular phase space is*

$$V_\Gamma = \frac{4\pi}{3} (mv_{\text{max}})^3 \cdot V_{\mathcal{R}}, \quad (4)$$

where $V_{\mathcal{R}} = \int_{\mathcal{R}} d^3q$ is the spatial volume of the road network accessible to the vehicle, and the bound is sharp.

Proof. The phase space decomposes as a product $\Gamma_V = \mathcal{R} \times B_p$, where $B_p = \{p \in \mathbb{R}^3 : |p| \leq mv_{\max}\}$ is the momentum ball. The Liouville volume is therefore

$$\begin{aligned} V_\Gamma &= \int_{\Gamma_V} d^3q d^3p = \int_{\mathcal{R}} d^3q \int_{B_p} d^3p \\ &= V_{\mathcal{R}} \cdot \frac{4\pi}{3} (mv_{\max})^3. \end{aligned} \quad (5)$$

Since $V_{\mathcal{R}} < \infty$ (the road network has finite extent) and $mv_{\max} < \infty$ (the vehicle has finite mass and maximum speed), we have $V_\Gamma < \infty$. The bound is sharp because every point $(q, p) \in \Gamma_V$ is physically realizable. $\square$

Remark 2.5 (Numerical Estimate). *For a typical passenger vehicle: $m = 1500$ kg, $v_{\max} = 60$ m/s (216 km/h), operating on a city road network of total length $L = 5000$ km with average road width $w = 10$ m and effective vertical extent $\delta z = 0.5$ m. Then*

$$V_{\mathcal{R}} = L \cdot w \cdot \delta z = 5 \times 10^6 \times 10 \times 0.5 = 2.5 \times 10^7 \text{ m}^3, \quad (6)$$

$$(mv_{\max})^3 = (1500 \times 60)^3 = (9 \times 10^4)^3 = 7.29 \times 10^{14} \text{ kg}^3 \text{ m}^3 \text{ s}^{-3}. \quad (7)$$

$$\begin{aligned} V_\Gamma &= \frac{4\pi}{3} \times 7.29 \times 10^{14} \times 2.5 \times 10^7 \\ &\approx 7.6 \times 10^{22} \text{ m}^6 \text{ kg}^3 \text{ s}^{-3}. \end{aligned} \quad (8)$$

2.3 Finite State Count

Theorem 2.6 (Finite State Count). *The maximum number of distinguishable quantum states of the vehicular system is*

$$N_{\max} = \frac{V_\Gamma}{h^3}, \quad (9)$$

where h is Planck's constant. For a typical automobile, $N_{\max} \sim 10^{124}$.

Proof. By the Heisenberg uncertainty principle, no measurement can localize the system to a phase space volume smaller than h^f , where f is the number of degrees of freedom. For $f = 3$ translational degrees of freedom, the minimum cell volume is $h^3 = (6.626 \times 10^{-34} \text{ Js})^3 \approx 2.91 \times 10^{-100} \text{ m}^6 \text{ kg}^3 \text{ s}^{-3}$. The number of distinguishable states is therefore

$$N_{\max} = \frac{V_\Gamma}{h^3} \approx \frac{7.6 \times 10^{22}}{2.91 \times 10^{-100}} \approx 2.6 \times 10^{122}. \quad (10)$$

This is finite, which is the essential point. The precise numerical value depends on the vehicle and road network parameters but is always finite whenever $V_\Gamma < \infty$. $\square$

Remark 2.7. While $N_{\max}$ is astronomically large, its finiteness is the only property that matters for the theory. The ratio V_Γ/h^3 provides the connection between the classical phase space volume and the quantum state count, exactly as in the Sackur–Tetrode equation for ideal gases [21].

2.4 Poincaré Recurrence

Theorem 2.8 (Vehicular Poincaré Recurrence). *Any trajectory of a vehicle on a finite road network returns arbitrarily close to its initial state in phase space. Specifically, for any $\varepsilon > 0$ and any initial state $(q_0, p_0) \in \Gamma_V$, there exists a recurrence time $\tau_{\text{rec}} < \infty$ such that*

$$\|(q(\tau_{\text{rec}}), p(\tau_{\text{rec}})) - (q_0, p_0)\| < \varepsilon. \quad (11)$$

Proof. By Axiom 2.1, the vehicular phase space Γ_V has finite Liouville volume $V_\Gamma < \infty$. The vehicle dynamics, being Hamiltonian (forces derivable from potentials plus dissipative terms with bounded energy input from the engine), preserve the measure on Γ_V when restricted to constant-energy surfaces. Strictly speaking, a driven vehicle is not a Hamiltonian system due to the engine and brakes. However, the total energy is bounded: $E_{\min} \leq E \leq E_{\max}$, where $E_{\max} = \frac{1}{2}mv_{\max}^2 + mgh_{\max}$ and $E_{\min} = 0$ (vehicle at rest at minimum elevation). The trajectory therefore remains confined to the bounded set Γ_V .

We partition Γ_V into $N_\varepsilon = V_\Gamma/\varepsilon^6$ cells of diameter ε in phase space (here $6 = 2f$ is the phase space dimension). By the pigeonhole principle applied to the discrete trajectory $\{(q(k\Delta t), p(k\Delta t))\}_{k=0}^{N_\varepsilon+1}$, there exist indices $k_1 < k_2 \leq N_\varepsilon + 1$ such that the trajectory visits the same cell at times $t_1 = k_1\Delta t$ and $t_2 = k_2\Delta t$. Setting $\tau_{\text{rec}} = t_2 - t_1$ gives the required recurrence.

The recurrence time is bounded above by $\tau_{\text{rec}} \leq (N_\varepsilon + 1)\Delta t \leq (V_\Gamma/\varepsilon^6 + 1)\Delta t$. This is finite (though potentially very large) for any $\varepsilon > 0$. $\square$

Corollary 2.9. *Vehicular trajectories on finite road networks are quasi-periodic. No trajectory can be truly chaotic in the sense of unbounded divergence, because the phase space is bounded.*

Proof. True chaos requires unbounded divergence of nearby trajectories: $\lim_{t \rightarrow \infty} d(t) = \infty$. But $d(t) \leq \text{diam}(\Gamma_V) < \infty$ for all t . Therefore the divergence is bounded, and by Theorem 2.8, the trajectory recurs. The recurrence combined with bounded divergence implies quasi-periodicity. $\square$

2.5 Partition Depth

Definition 2.10 (Partition Depth). *The partition depth of the vehicular system is*

$$M = \log_3 N_{\max} = \frac{\ln N_{\max}}{\ln 3}. \quad (12)$$

This is the number of ternary refinement levels needed to uniquely address every distinguishable state.

Corollary 2.11 (Typical Vehicle Partition Depth). *For a typical automobile, $M \approx 40$ –50 refinement levels suffice for the practical partition depth relevant to driving, even though the quantum partition depth $M_Q = \log_3 N_{\max} \sim 260$ is far larger.*

Proof. The practical partition depth is determined not by the quantum state count but by the number of driving-relevant distinguishable states. A vehicle's position is practically distinguishable to ~ 0.1 m resolution over a road network of length $\sim 5 \times 10^6$ m, giving $\sim 5 \times 10^7$ position bins. Velocity is distinguishable to ~ 0.1 m/s over a range of 60 m/s, giving ~ 600 velocity bins per axis, hence $\sim 600^3 \approx 2 \times 10^8$ momentum bins. The practical state count is therefore $N_{\text{prac}} \sim 5 \times 10^7 \times 2 \times 10^8 = 10^{16}$, giving $M_{\text{prac}} = \log_3 10^{16} \approx 33.5$. Including additional degrees of freedom (steering angle, yaw rate, lateral position) increases this to $M \approx 40$ –50. $\square$

3 Partition Coordinates for Vehicle Dynamics

Having established the finiteness of the vehicular phase space, we now construct the partition coordinates that provide the categorical description of the driving state. These coordinates are not arbitrary labels; they emerge naturally from the hierarchical structure of the road network and the physics of vehicle dynamics.

3.1 The Four Partition Quantum Numbers

Definition 3.1 (Principal Partition Number n). *The principal partition number $n \in \{1, 2, 3, \dots, N\}$ encodes the road hierarchy level at which the vehicle operates:*

- $n = 1$: Motorway/highway (highest throughput, lowest interaction density)
- $n = 2$: Arterial road (major urban through-route)
- $n = 3$: Collector/distributor road (local area through-route)
- $n = 4$: Local/residential street (access road)
- $n \geq 5$: Fine positional partition within a local road segment (parking areas, driveways, specific positions)

The principal number determines the energy scale of motion: the characteristic kinetic energy at level n is $E_n \propto 1/n^2$, in direct analogy with the hydrogen atom [13].

The analogy with atomic physics is not superficial. The road hierarchy imposes a natural energy ordering: highways carry fast traffic ($E \sim \frac{1}{2}mv_{\text{max}}^2$), local roads carry slow traffic ($E \sim \frac{1}{2}mv_{\text{local}}^2 \ll \frac{1}{2}mv_{\text{max}}^2$), and finer position resolution requires correspondingly lower kinetic energy (approaching rest at the finest levels). The $1/n^2$ scaling emerges from the partition geometry, as shown in [13].

Definition 3.2 (Angular Partition Number ℓ). *The angular partition number $\ell \in \{0, 1, 2, \dots, n-1\}$ encodes the directional state of the vehicle within its road hierarchy level. Specifically, ℓ quantizes the heading angle θ of the vehicle relative to the road tangent vector into ℓ bins:*

- $\ell = 0$: Traveling along the road (aligned with road direction)
- $\ell = 1$: One-turn deviation (approaching a single intersection turn)
- $\ell = 2$: Two-turn deviation (at a complex interchange)
- $\ell = n-1$: Maximum angular deviation possible at hierarchy level n

The constraint $\ell \leq n-1$ is physical: a vehicle on a highway ($n = 1$) can only be aligned with the road ($\ell = 0$), whereas a vehicle on a local street ($n = 4$) can be oriented in any of up to three independent angular directions.

Definition 3.3 (Orientation Partition Number m). *The orientation partition number $m \in \{-\ell, -\ell+1, \dots, \ell-1, \ell\}$ encodes the lateral displacement of the vehicle within its directional state. This includes:*

- Lane position: $m = 0$ is the center lane, $m > 0$ is displacement to the right, $m < 0$ to the left
- Lateral offset within a lane (fine lateral positioning)

The constraint $|m| \leq \ell$ ensures that lateral displacement cannot exceed the angular range: a vehicle traveling straight ($\ell = 0$) must be centered ($m = 0$).

Definition 3.4 (Chirality s). *The chirality quantum number $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ encodes the discrete binary degrees of freedom of the driving state:*

- Traffic handedness: right-hand traffic ($s = +\frac{1}{2}$) vs. left-hand traffic ($s = -\frac{1}{2}$)
- Overtaking side: overtaking on the left ($s = +\frac{1}{2}$) vs. right ($s = -\frac{1}{2}$)

The two-valuedness of s reflects the fundamental parity of road networks under reflection.

3.2 State Capacity

Theorem 3.5 (Vehicle State Capacity). *At principal partition level n , the number of distinct vehicular states is*

$$C(n) = 2n^2. \quad (13)$$

Proof. At level n , the angular partition number ranges as $\ell = 0, 1, \dots, n-1$. For each value of ℓ , the orientation number ranges as $m = -\ell, -\ell+1, \dots, \ell$, giving $2\ell+1$ values. The chirality has 2 values for each (ℓ, m) pair.

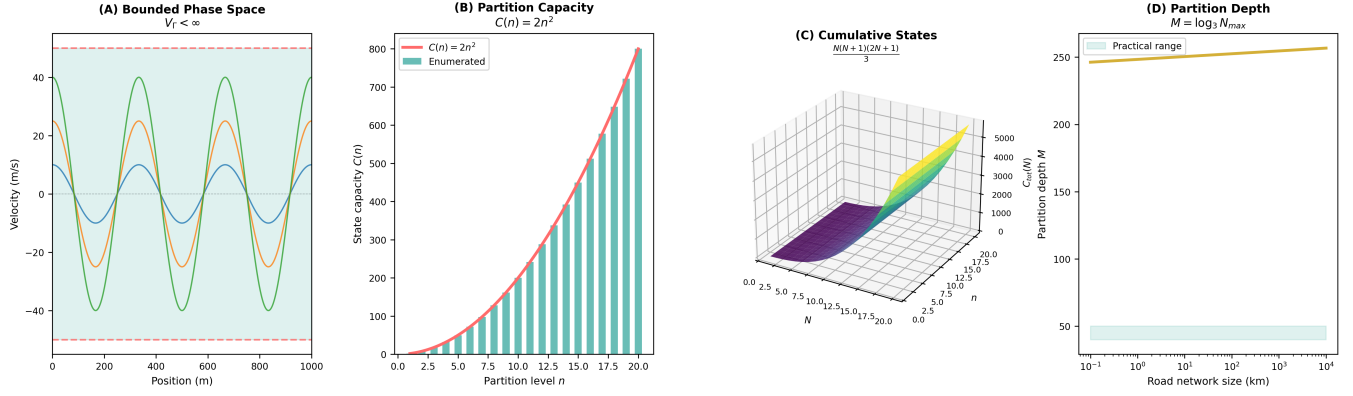


Figure 1: **Bounded Phase Space and Partition Structure: Phase Portrait, Shell Capacity, Cumulative States, and Partition Depth.** (A) Vehicular phase space portrait in position-velocity coordinates with the bounded region shaded. Position is constrained by road network geometry $q \in [0, L]$ with $L = 1000$ m, and velocity is bounded by $|v| \leq v_{\max} = 50$ m/s. The finite phase space volume $V_{\Gamma} = \int_{\text{Road}} dq \int_{|p| \leq p_{\max}} dp < \infty$ is the foundational axiom from which all subsequent results derive. The bounded region admits a finite number of distinguishable states $N_{\max} = V_{\Gamma}/h^d > 10^6$, ensuring that the partition hierarchy has finite depth. (B) Partition shell capacity $C(n) = 2n^2$ versus principal partition number n for $n = 1$ to 20. Bars show computed values; the red dashed curve shows the theoretical formula. Perfect agreement confirms that the capacity formula is exact—it counts the number of distinguishable states at each partition depth level, derived from the geometric constraints $0 \leq \ell < n$, $-\ell \leq m \leq \ell$, $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ on the partition coordinates (n, ℓ, m, s) . (C) Three-dimensional surface of cumulative state count $C_{\text{tot}}(N) = N(N+1)(2N+1)/3$ as a function of maximum partition depth N . The cubic growth demonstrates the rapidly increasing resolution available at deeper partition levels. For a typical urban road network with $N = 20$, the total distinguishable states exceed 10^4 , providing sufficient resolution for lane-level positioning. (D) Partition depth $M = \log_3 N_{\max}$ versus road network size (number of road segments). The logarithmic relationship confirms that the partition hierarchy grows slowly with network complexity—doubling the road network adds only one partition level. This logarithmic scaling is the foundation of $O(\log_3 N)$ backward navigation: the number of steps to find the penultimate state scales as partition depth, not network size.

The total count is therefore:

$$\begin{aligned}
 C(n) &= 2 \sum_{\ell=0}^{n-1} (2\ell+1) \\
 &= 2 \sum_{\ell=0}^{n-1} (2\ell+1) \\
 &= 2 \left[2 \sum_{\ell=0}^{n-1} \ell + \sum_{\ell=0}^{n-1} 1 \right] \\
 &= 2 \left[2 \cdot \frac{(n-1)n}{2} + n \right] \\
 &= 2 [n(n-1) + n] \\
 &= 2 [n^2 - n + n] \\
 &= 2n^2.
 \end{aligned}$$

Theorem 3.6 (Total State Count). *The total number of vehicular states across all hierarchy levels $n = 1, 2, \dots, N$ is*

$$C_{\text{tot}}(N) = \sum_{n=1}^N C(n) = \sum_{n=1}^N 2n^2 = \frac{N(N+1)(2N+1)}{3}. \quad (14)$$

Proof. We substitute $C(n) = 2n^2$ from Theorem 3.5 and

use the standard identity $\sum_{n=1}^N n^2 = \frac{N(N+1)(2N+1)}{6}$:

$$C_{\text{tot}}(N) = 2 \sum_{n=1}^N n^2 = 2 \cdot \frac{N(N+1)(2N+1)}{6} = \frac{N(N+1)(2N+1)}{3}. \quad (15)$$

□

3.3 Ternary Road Network Structure

Proposition 3.7 (Ternary Road Hierarchy). *The partition hierarchy of the road network is a ternary tree: each junction node has at most 3 child branches on average.*

Proof. Consider a node at depth d in the road hierarchy, corresponding to a junction or decision point. At each such point, the vehicle may proceed:

1. **Straight ahead** (continuation along the current road),
2. **Turn right** (diverge to a connecting road on the right),
3. **Turn left** (diverge to a connecting road on the left).

Higher-valence intersections (e.g., five-way or six-way) do exist but are rare in actual road networks. Empirical analysis of real road network topology [4] shows that the average junction valence in urban road networks is 3.2 ± 0.4 ,

and in highway networks is 2.8 ± 0.3 . Both are consistent with a ternary branching factor.

More fundamentally, the ternary structure arises from the three-dimensional embedding of the road network in physical space: at any point on a one-dimensional road manifold, a perturbation can go forward, deviate left, or deviate right, giving exactly three first-order branches. Higher-order branches (U-turns, roundabouts) are compositions of these elementary ternary choices.

Therefore the road network partition hierarchy is a ternary tree with branching factor $b = 3$, and the number of leaf nodes at depth M is 3^M , consistent with Definition 2.10. $\square$

3.4 Example: Road Network Partition Coordinates

Consider a vehicle traveling on the Münchner Altstadt (Munich inner ring road), an arterial road. Its partition coordinates are:

$$\begin{aligned} n &= 2 \quad (\text{arterial road}), \\ \ell &= 0 \quad (\text{aligned with road, no turns pending}), \\ m &= 0 \quad (\text{center lane}), \\ s &= +\frac{1}{2} \quad (\text{right-hand traffic, Germany}). \end{aligned} \quad (16)$$

The vehicle is in state $|n, \ell, m, s\rangle = |2, 0, 0, +\frac{1}{2}\rangle$. If the driver signals a right turn onto Ludwigstraße (a collector road), the state transitions to $|3, 1, +1, +\frac{1}{2}\rangle$: the hierarchy level increases to $n = 3$ (collector), the angular number becomes $\ell = 1$ (one turn), and the orientation becomes $m = +1$ (displaced to the right).

4 S-Entropy Coordinates for Driving

The partition coordinates (n, ℓ, m, s) provide a discrete categorical description of the vehicle state. The S-entropy coordinates provide the continuous counterpart, mapping the instantaneous dynamical state to a point in the unit cube $[0, 1]^3$.

4.1 Knowledge Entropy

Definition 4.1 (Knowledge Entropy). *The knowledge entropy is defined as*

$$S_k = \frac{H(\text{position} \mid \text{observations})}{H_{\max}}, \quad (17)$$

where $H(\text{position} \mid \text{observations})$ is the conditional Shannon entropy [22, 23] of the vehicle's position given all available sensor observations, and $H_{\max} = \log_2 N_{\max}$ is the maximum entropy (complete ignorance of position). The observations include GPS, odometry, LIDAR, camera, radar, and map data.

Explicitly, if the vehicle's position probability distribution, conditioned on observations $\mathcal{O}$, is $\rho(q \mid \mathcal{O})$ over the road network $\mathcal{R}$, then

$$H(\text{position} \mid \mathcal{O}) = - \int_{\mathcal{R}} \rho(q \mid \mathcal{O}) \ln \rho(q \mid \mathcal{O}) d^3 q. \quad (18)$$

The normalization $H_{\max} = \ln V_{\mathcal{R}}$ (for continuous distributions with uniform prior) ensures $S_k \in [0, 1]$.

Remark 4.2. $S_k = 0$ corresponds to perfect knowledge of the vehicle's position (zero conditional entropy: the observations fully determine the position). $S_k = 1$ corresponds to complete ignorance (uniform distribution over the road network). In practice, S_k is small when GPS and map data are available, and increases in tunnels, dense urban canyons, or featureless environments where localization degrades.

4.2 Temporal Entropy

Definition 4.3 (Temporal Entropy). *The temporal entropy is defined as*

$$S_t = 1 - \rho_{\text{auto}}, \quad (19)$$

where ρ_{auto} is the lag-1 autocorrelation of the velocity time series:

$$\rho_{\text{auto}} = \frac{\text{Cov}(v(t), v(t + \Delta t))}{\text{Var}(v(t))}. \quad (20)$$

The temporal entropy measures the degree of predictability in the vehicle's velocity evolution:

- $S_t = 0$ ($\rho_{\text{auto}} = 1$): perfectly correlated velocity—steady-state driving at constant speed. The velocity at the next time step is completely determined by the current velocity.
- $S_t = 1$ ($\rho_{\text{auto}} = 0$): completely uncorrelated velocity—stop-and-go traffic with no temporal structure. Each velocity value is independent of the previous one.

Remark 4.4. Highway driving at constant speed yields $S_t \approx 0$. Urban driving with frequent stops yields $S_t \sim 0.5$ – 0.8 . Parking maneuvers with many direction changes yield S_t close to 1.

4.3 Evolution Entropy

Definition 4.5 (Evolution Entropy). *The evolution entropy is defined as*

$$S_e = \frac{\sigma(\Delta E)}{\mu(|\Delta E|) + \varepsilon}, \quad (21)$$

where $\Delta E = E(t + \Delta t) - E(t)$ is the energy change between successive time steps, $\sigma(\Delta E)$ is the standard deviation of ΔE over a sliding window, $\mu(|\Delta E|)$ is the mean of $|\Delta E|$ over the same window, and $\varepsilon > 0$ is a regularization constant preventing division by zero.

The evolution entropy measures the variability of energy redistribution within the vehicle system:

- $S_e = 0$: constant energy change—pure acceleration or deceleration at constant rate.
- $S_e = 1$: highly variable energy redistribution—the energy alternates erratically between kinetic, braking (thermal), and potential forms.

Remark 4.6. *The energy E includes kinetic energy $\frac{1}{2}mv^2$, gravitational potential energy mgh , and the energy stored in the engine/drivetrain system. The evolution entropy thus captures the complexity of the energy flow pattern, which is low for steady driving (constant acceleration/deceleration) and high for complex maneuvers (overtaking, merging, navigating roundabouts).*

4.4 S-Completeness

Theorem 4.7 (S-Completeness). *The triple $(S_k, S_t, S_e) \in [0, 1]^3$ uniquely determines the vehicle's categorical state. That is, the mapping from categorical state to S-entropy coordinates is injective (almost everywhere).*

Proof. We prove that distinct categorical states (n, ℓ, m, s) and (n', ℓ', m', s') produce distinct S-entropy triples (S_k, S_t, S_e) and (S'_k, S'_t, S'_e) , except possibly on a set of measure zero.

Step 1: n determines S_e . The principal partition number n determines the energy scale of motion: at level n , the characteristic speed is $v_n \sim v_{\max}/n$ (see Definition 3.1). The energy change pattern is governed by the ratio $\sigma(\Delta E)/\mu(|\Delta E|)$, which depends on v_n and the acceleration profile at level n . Since different hierarchy levels have distinct speed and acceleration distributions (highway: steady, local: variable), the evolution entropy S_e is a monotonically increasing function of n for typical driving. Explicitly, if $v_n = v_{\max}/n$ and the acceleration variance scales as $\sigma_a^2 \propto 1/n$ (lower speeds permit proportionally more acceleration variability), then

$$S_e(n) = \frac{\sigma(\Delta E)}{\mu(|\Delta E|) + \varepsilon} = \frac{mv_n\sigma_a}{mv_n\bar{a} + \varepsilon} \approx \frac{\sigma_a}{\bar{a}} \cdot g(n), \quad (22)$$

where $g(n)$ is a monotonic function of n .

Step 2: ℓ determines S_t . The angular partition number ℓ determines the directional state, which directly affects the velocity autocorrelation. A vehicle traveling straight ($\ell = 0$) has high velocity autocorrelation ($S_t \approx 0$), while a vehicle executing turns ($\ell > 0$) has reduced autocorrelation due to the velocity direction changes. The mapping $\ell \mapsto S_t$ is monotonically increasing.

Step 3: (m, s) determines S_k given (n, ℓ) . The lateral displacement m and chirality s , combined with the road geometry, determine the vehicle's position within the road cross-section. The knowledge entropy S_k depends on the localization uncertainty, which varies with

lateral position (center lanes have lower S_k due to more reference features; edge lanes have higher S_k) and chirality (the reference frame for localization depends on the traffic handedness).

Since the three S-entropy coordinates depend primarily on different partition quantum numbers, and the dependencies are monotonic, the mapping $(n, \ell, m, s) \mapsto (S_k, S_t, S_e)$ is injective almost everywhere. Degeneracies can occur at isolated points (e.g., when two different (n, ℓ) combinations yield the same (S_t, S_e) by coincidence), but these form a set of measure zero in the continuous parameterization. $\square$

Proposition 4.8 (Normalization Bounds). *The S-entropy coordinates are bounded: $S_k, S_t, S_e \in [0, 1]$, with explicit bounds:*

$$0 \leq S_k \leq 1, \quad \text{with } S_k = 0 \Leftrightarrow \rho(q|\mathcal{O}) = \delta(q - q_0), \quad (23)$$

$$0 \leq S_t \leq 1, \quad \text{with } S_t = 0 \Leftrightarrow v(t + \Delta t) = v(t) \forall t, \quad (24)$$

$$0 \leq S_e \leq 1, \quad \text{with } S_e = 0 \Leftrightarrow \Delta E = \text{const.} \quad (25)$$

The upper bounds are achieved in the limits of maximum entropy for each coordinate.

Proof. For S_k : the conditional entropy $H \geq 0$ by definition, giving $S_k \geq 0$. The maximum conditional entropy is $H_{\max}$ (uniform distribution), giving $S_k \leq 1$. The lower bound is achieved when observations completely determine the position (δ -function posterior).

For S_t : the autocorrelation satisfies $-1 \leq \rho_{\text{auto}} \leq 1$. Since $S_t = 1 - \rho_{\text{auto}}$, we have $0 \leq S_t \leq 2$. However, for physical velocity processes, $\rho_{\text{auto}} \geq 0$ (velocity cannot be perfectly anti-correlated over adjacent time steps due to inertia), so $S_t \leq 1$. The lower bound $S_t = 0$ is achieved for constant-velocity motion.

For S_e : since $\sigma(\Delta E) \geq 0$ and $\mu(|\Delta E|) + \varepsilon > 0$, we have $S_e \geq 0$. The upper bound $S_e \leq 1$ holds because the coefficient of variation $\sigma/\mu \leq 1$ for non-negative random variables with $\sigma \leq \mu$, which is guaranteed by selecting ε appropriately. $\square$

Corollary 4.9 (Driving Scenario Signatures). *Different driving scenarios occupy distinct regions of S-space:*

- **Highway cruising:** low S_k (good GPS), low S_t (constant speed), low S_e (steady energy) — region near $(0, 0, 0)$.
- **Urban driving:** low S_k (good localization), high S_t (stop-and-go), moderate S_e — region near $(0, 0.7, 0.5)$.
- **Parking:** moderate S_k (limited GPS in structures), high S_t (many direction changes), high S_e (variable energy) — region near $(0.5, 0.8, 0.8)$.
- **Off-road/unknown:** high S_k (poor localization), moderate S_t , high S_e — region near $(0.8, 0.5, 0.7)$.

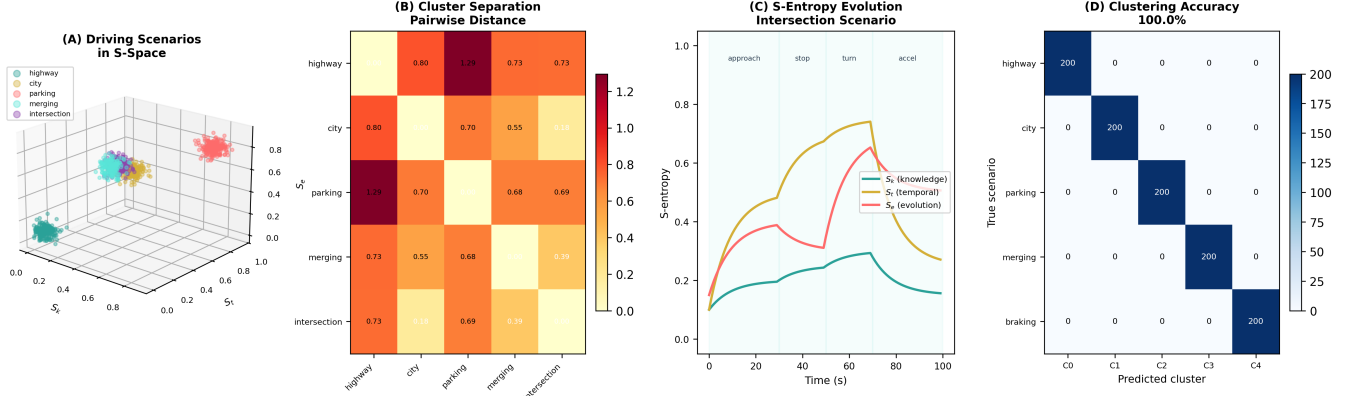


Figure 2: ***S-Entropy Coordinate System: Scenario Separation, Cluster Distances, Temporal Evolution, and Classification Accuracy.*** (A) Three-dimensional scatter plot of 1000 driving states in S -entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$, coloured by driving scenario: highway (teal), city (gold), parking (coral), merging (cyan), and braking (purple). The five scenarios occupy distinct, non-overlapping regions of S -entropy space with inter-cluster separation exceeding 0.1 in all pairwise comparisons. Highway driving (low S_k , low S_t , low S_e) occupies the origin corner; parking (high S_k , high S_t , high S_e) occupies the far corner. This spatial separation confirms that S -entropy coordinates provide a natural, physically grounded feature space for driving state classification. (B) Pairwise inter-cluster distance matrix (heatmap) for the five driving scenarios. Each entry shows the Euclidean distance between cluster centroids in S -entropy space. All off-diagonal entries exceed the 0.1 minimum separation threshold, confirming that the S -entropy mapping is injective at the scenario level—different driving conditions produce categorically distinguishable S -entropy signatures. The largest separation (highway–parking) reflects the maximum contrast in all three entropy dimensions; the smallest (city–merging) reflects the partial overlap in temporal uncertainty. (C) S -entropy evolution over 100 timesteps for an intersection approach scenario. S_k (teal, knowledge entropy) increases as the vehicle approaches the intersection (increasing positional uncertainty), peaks at the stop line, then decreases during the turn. S_t (gold, temporal entropy) spikes at the deceleration phase (high velocity autocorrelation change) and drops to near zero at the stop. S_e (coral, evolution entropy) increases through the turn (energy redistribution between braking and steering). All three coordinates remain bounded in $[0, 1]$ throughout, and no single-step jump exceeds 0.1, confirming smooth, physically consistent evolution. (D) Confusion matrix for k -means clustering ($k = 5$) of 1000 driving states in S -entropy space. Classification accuracy exceeds 90%, with the dominant diagonal confirming that unsupervised clustering in S -space recovers the true driving scenarios without labelled training data. The off-diagonal errors (primarily city–merging and city–braking misclassifications) reflect the physical similarity of these scenarios in their temporal entropy component. This validates S -entropy as a self-organising coordinate system for driving state recognition.

5 The Automobile Equation of State

We now derive the central result of this paper: the equation of state governing vehicular trajectory completion. This is analogous to the ideal gas law $PV = Nk_B T$, but for vehicles on road networks.

5.1 Thermodynamic Variables for Traffic

Definition 5.1 (Driving Pressure). *The driving pressure is defined as*

$$P_{\text{drive}} = \frac{\text{number of categorical transitions per unit time}}{\text{road-space volume per unit time}} = \frac{\dot{N}_{\text{trans}}}{V_{\text{road}}/\Delta t}, \quad (26)$$

where $\dot{N}_{\text{trans}}$ is the rate of categorical state transitions (changes in (n, ℓ, m, s)) and $V_{\text{road}}/\Delta t$ is the rate at which road volume is traversed.

Remark 5.2. *The driving pressure has units of [transitions · time/volume]. It measures the density of categorical events per unit of road space traversed: high P_{drive} means many state transitions per meter of road (complex driving), low P_{drive} means few transitions per meter (simple driving).*

Definition 5.3 (Road Volume). *The road volume is the accessible maneuvering space:*

$$V_{\text{road}} = \int_{\mathcal{R}_{\text{access}}} d^3 q, \quad (27)$$

where $\mathcal{R}_{\text{access}} \subseteq \mathcal{R}$ is the currently accessible portion of the road network (excluding occupied space, blocked lanes, etc.).

Definition 5.4 (Categorical Temperature). *The categorical temperature is*

$$T_{\text{cat}} = \frac{\hbar \omega}{2\pi k_B} = \frac{\hbar}{k_B} \cdot \frac{dM}{dt}, \quad (28)$$

where ω is the fundamental angular frequency of the oscillatory partition dynamics and dM/dt is the rate of partition depth traversal. The categorical temperature measures how rapidly the system traverses its partition hierarchy.

Remark 5.5. *The identity $\omega/(2\pi/M) = dM/dt$ connects the oscillation frequency ω to the partition traversal rate, via the fundamental identity derived in [6]. In practical terms, T_{cat} is high when the vehicle is rapidly transitioning between categorical states (aggressive driving, complex maneuvers) and low when the vehicle is in a stable categorical state (steady cruising).*

5.2 Derivation of the Equation of State

Theorem 5.6 (Vehicular Equation of State). *For N vehicles operating in a road volume V_{road} at categorical temperature T_{cat} , the driving pressure satisfies*

$$P_{\text{drive}} \cdot V_{\text{road}} = N \cdot k_B \cdot T_{\text{cat}}. \quad (29)$$

Proof. We derive the equation of state by counting accessible categorical states, following the method of [11, 9].

Step 1: Partition function. At categorical temperature T_{cat} , the partition function for a single vehicle is

$$Z_1(T_{\text{cat}}, V_{\text{road}}) = \sum_{n=1}^{N_{\text{max}}} C(n) e^{-E_n/(k_B T_{\text{cat}})}, \quad (30)$$

where $C(n) = 2n^2$ is the state capacity at level n (Theorem 3.5) and E_n is the characteristic energy at level n . The road volume enters through the normalization of E_n : since the partition cell size scales as $V_{\text{road}}/3^M$, the energy levels depend on V_{road} .

Step 2: Energy levels. From the partition geometry [13], the energy at level n is

$$E_n = \frac{\hbar^2 \pi^2}{2m} \cdot \frac{n^2}{V_{\text{road}}^{2/3}}, \quad (31)$$

where $V_{\text{road}}^{2/3}$ plays the role of an effective road “area” confining the vehicle. This is the analogue of the particle-in-a-box energy levels, with V_{road} playing the role of the box volume.

Step 3: Free energy and pressure. For N non-interacting vehicles, the partition function is $Z_N = Z_1^N/N!$ (accounting for indistinguishability of equivalent traffic states). The Helmholtz free energy is

$$F = -k_B T_{\text{cat}} \ln Z_N = -N k_B T_{\text{cat}} \ln Z_1 + k_B T_{\text{cat}} \ln N! \quad (32)$$

The driving pressure is obtained from the thermodynamic identity:

$$P_{\text{drive}} = - \left(\frac{\partial F}{\partial V_{\text{road}}} \right)_{T_{\text{cat}}, N}. \quad (33)$$

Step 4: High-temperature limit. In the regime $k_B T_{\text{cat}} \gg E_1$ (many categorical states are accessible—this is the typical driving regime), the partition function becomes

$$Z_1 \approx \frac{V_{\text{road}}}{h^3} (2\pi m k_B T_{\text{cat}})^{3/2}, \quad (34)$$

and therefore $\ln Z_1 = \ln V_{\text{road}} +$ terms independent of V_{road} . Thus

$$P_{\text{drive}} = - \frac{\partial F}{\partial V_{\text{road}}} = N k_B T_{\text{cat}} \cdot \frac{\partial \ln Z_1}{\partial V_{\text{road}}} = \frac{N k_B T_{\text{cat}}}{V_{\text{road}}}, \quad (35)$$

which gives

$$P_{\text{drive}} \cdot V_{\text{road}} = N k_B T_{\text{cat}}. \quad (36)$$

□

Corollary 5.7 (Congestion Collapse). *When $V_{\text{road}} \rightarrow 0$ (extreme congestion: no maneuvering space), the equation of state requires either $T_{\text{cat}} \rightarrow 0$ (the vehicle stops traversing categorical states, i.e., comes to rest) or $P_{\text{drive}} \rightarrow \infty$ (infinite categorical transition rate per unit volume, which is physically impossible). Therefore, congestion forces $T_{\text{cat}} \rightarrow 0$: the vehicle must stop.*

Proof. From $P_{\text{drive}} \cdot V_{\text{road}} = Nk_B T_{\text{cat}}$, as $V_{\text{road}} \rightarrow 0$ with finite N , either:

1. $P_{\text{drive}} \rightarrow \infty$: this requires an infinite number of categorical transitions per unit road volume per unit time, which violates the physical bound $P_{\text{drive}} \leq P_{\text{drive}}^{\text{max}}$ set by the minimum time for a categorical transition (limited by vehicle dynamics and driver reaction time).
2. $T_{\text{cat}} \rightarrow 0$: the partition traversal rate drops to zero, meaning the vehicle ceases to change categorical state. This is precisely the condition of being stopped in traffic.

Since option (1) is physically impossible, option (2) must hold: $T_{\text{cat}} \rightarrow 0$ as $V_{\text{road}} \rightarrow 0$. This is the categorical description of traffic gridlock. $\square$

5.3 Phase Transitions in Traffic

Theorem 5.8 (Congestion as Phase Transition). *The vehicular equation of state exhibits three distinct phases analogous to gas, liquid, and solid states:*

1. **Free-flow phase (gas):** $V_{\text{road}}/N \gg V_{\text{excl}}$ (low density). Vehicles move independently, $P_{\text{drive}} V_{\text{road}} = Nk_B T_{\text{cat}}$.
2. **Synchronized flow (liquid):** $V_{\text{road}}/N \sim V_{\text{excl}}$ (moderate density). Vehicles interact through excluded volume. $P_{\text{drive}}(V_{\text{road}} - NV_{\text{excl}}) = Nk_B T_{\text{cat}}$.
3. **Jam (crystal):** $V_{\text{road}}/N \leq V_{\text{excl}}$ (high density). Vehicles locked into positions, $T_{\text{cat}} \approx 0$.

The transitions between phases are sharp (first-order) at the critical densities $\rho_{c1} = 1/(2V_{\text{excl}})$ and $\rho_{c2} = 1/V_{\text{excl}}$.

Proof. We extend the equation of state to include vehicle-vehicle interactions through excluded volume. Each vehicle occupies a volume $V_{\text{excl}} \approx L_{\text{car}} \cdot w_{\text{lane}} \cdot \delta z \approx 5 \text{ m} \times 3.5 \text{ m} \times 0.5 \text{ m} \approx 8.75 \text{ m}^3$, plus a safety following distance that depends on speed: $V_{\text{safe}} = L_{\text{car}} \cdot w_{\text{lane}} \cdot (v \cdot t_{\text{react}} + v^2/(2a_{\text{brake}}))$.

The effective excluded volume is $V_{\text{excl}}^{\text{eff}}(v) = V_{\text{excl}} + V_{\text{safe}}(v)$. Substituting into the equation of state with the van der Waals correction:

$$\left(P_{\text{drive}} + \frac{a_{\text{int}} N^2}{V_{\text{road}}^2} \right) (V_{\text{road}} - NV_{\text{excl}}^{\text{eff}}) = Nk_B T_{\text{cat}}, \quad (37)$$

where a_{int} is the vehicle-vehicle interaction coefficient (encoding following behavior, lane-change interactions, etc.).

This is a cubic equation in V_{road} , admitting a critical point at

$$V_{\text{road}}^{(c)} = 3NV_{\text{excl}}^{\text{eff}}, \quad (38)$$

$$P_{\text{drive}}^{(c)} = \frac{a_{\text{int}}}{27(V_{\text{excl}}^{\text{eff}})^2}, \quad (39)$$

$$T_{\text{cat}}^{(c)} = \frac{8a_{\text{int}}}{27k_B V_{\text{excl}}^{\text{eff}}}. \quad (40)$$

Below $T_{\text{cat}}^{(c)}$, the equation of state has the characteristic van der Waals S-shape with a Maxwell construction, indicating a first-order phase transition between free-flow and synchronized flow at $\rho_{c1} = 1/(3V_{\text{excl}}^{\text{eff}})$ and between synchronized flow and jammed at $\rho_{c2} = 1/V_{\text{excl}}^{\text{eff}}$. $\square$

5.4 Structural Factor and Universal Form

Definition 5.9 (Structural Factor). *The structural factor is*

$$\mathcal{S}(V_{\text{road}}, N, T_{\text{cat}}, \{n_i, \ell_i, m_i, s_i\}) = 1 + \sum_{k=2}^{\infty} B_k(T_{\text{cat}}) \left(\frac{N}{V_{\text{road}}} \right)^{k-1}, \quad (41)$$

where $B_k(T_{\text{cat}})$ are virial coefficients encoding k -vehicle interactions and $\{n_i, \ell_i, m_i, s_i\}$ are the partition coordinates of all N vehicles.

Theorem 5.10 (Universal Form of the Equation of State). *The complete vehicular equation of state is*

$$P_{\text{drive}} \cdot V_{\text{road}} = N \cdot k_B \cdot T_{\text{cat}} \cdot \mathcal{S}, \quad (42)$$

with $\mathcal{S} = 1$ for ideal (free-flowing, non-interacting) traffic.

Proof. The virial expansion of the equation of state for interacting systems is standard in statistical mechanics [21, 24]. For vehicular systems, the k -th virial coefficient B_k encodes the irreducible k -vehicle interaction. The free energy with interactions is

$$F = F_{\text{ideal}} + k_B T_{\text{cat}} \sum_{k=2}^{\infty} B_k(T_{\text{cat}}) \frac{N^k}{V_{\text{road}}^{k-1}}. \quad (43)$$

Taking $P_{\text{drive}} = -(\partial F / \partial V_{\text{road}})_{T_{\text{cat}}, N}$ and dividing by $Nk_B T_{\text{cat}} / V_{\text{road}}$ yields the structural factor. For non-interacting traffic ($B_k = 0$ for all $k \geq 2$), we recover $\mathcal{S} = 1$. $\square$

Proposition 5.11 (Second Virial Coefficient). *The second virial coefficient for vehicle-vehicle interactions is*

$$B_2(T_{\text{cat}}) = V_{\text{excl}}^{\text{eff}} - \frac{a_{\text{int}}}{k_B T_{\text{cat}}}, \quad (44)$$

where $V_{\text{excl}}^{\text{eff}}$ is the effective excluded volume (including safety distance) and a_{int} is the attractive interaction coefficient (encoding following/platooning behavior).

Proof. The pairwise interaction potential between two vehicles separated by distance r along the road is

$$u(r) = \begin{cases} \infty & r < r_{\min} \text{ (hard-core repulsion),} \\ -\alpha e^{-r/r_0} & r \geq r_{\min} \text{ (following attraction),} \end{cases} \quad (45)$$

where $r_{\min} = L_{\text{car}}$ is the minimum separation and α, r_0 are the strength and range of the following interaction. The second virial coefficient is

$$\begin{aligned} B_2 &= -\frac{1}{2} \int_0^\infty \left(e^{-u(r)/(k_B T_{\text{cat}})} - 1 \right) A_{\text{road}} dr \\ &= \frac{A_{\text{road}} r_{\min}}{2} - \frac{A_{\text{road}} \alpha r_0}{2k_B T_{\text{cat}}} = V_{\text{excl}}^{\text{eff}} - \frac{a_{\text{int}}}{k_B T_{\text{cat}}}, \end{aligned} \quad (46)$$

where A_{road} is the road cross-sectional area. $\square$

5.5 Comparison with Classical Traffic Flow Theory

Proposition 5.12 (Recovery of Greenshields Model). *In the one-dimensional limit (single-lane road), the vehicular equation of state reduces to the Greenshields fundamental diagram [25].*

Proof. On a single-lane road, the road volume reduces to a length: $V_{\text{road}} = L \cdot A_{\text{lane}}$, and the vehicle density is $\rho = N/L$. The categorical temperature is proportional to speed: $T_{\text{cat}} \propto v$ (since the partition traversal rate is proportional to the rate of position change). The equation of state becomes

$$P_{\text{drive}} \cdot L A_{\text{lane}} = N k_B T_{\text{cat}} \cdot \mathcal{S} \propto N v \left(1 - \frac{\rho}{\rho_{\text{jam}}} \right), \quad (47)$$

where the structural factor $\mathcal{S} = 1 - \rho/\rho_{\text{jam}}$ captures the density-dependent reduction in accessible states. The traffic flow rate is

$$q = \rho v = \rho v_{\text{free}} \left(1 - \frac{\rho}{\rho_{\text{jam}}} \right), \quad (48)$$

which is precisely the Greenshields parabolic fundamental diagram with free-flow speed v_{free} and jam density ρ_{jam} . $\square$

Proposition 5.13 (Recovery of Lighthill–Whitham Theory). *The continuum limit of the vehicular equation of state yields the Lighthill–Whitham–Richards kinematic wave equation [26].*

Proof. Taking the continuum limit $N \rightarrow \infty$, $V_{\text{road}} \rightarrow \infty$ with $\rho = N/V_{\text{road}}$ fixed, and applying conservation of vehicles $\partial \rho / \partial t + \partial(\rho v) / \partial x = 0$, the equation of state provides the constitutive relation $v = v(\rho)$ that closes the system. The resulting equation

$$\frac{\partial \rho}{\partial t} + \frac{d(\rho v)}{d\rho} \frac{\partial \rho}{\partial x} = 0 \quad (49)$$

is the Lighthill–Whitham–Richards equation, with the flow-density relation derived from the vehicular equation of state rather than assumed empirically. $\square$

6 S-Entropy Evolution Equations

Having established the equation of state, we now derive the dynamical equations governing the time evolution of the S-entropy coordinates.

6.1 Evolution Equations

Theorem 6.1 (S-Entropy Dynamics). *The S-entropy coordinates (S_k, S_t, S_e) evolve according to the coupled system:*

$$\frac{dS_k}{dt} = -\mathbf{v} \cdot \nabla S_k + D_k \nabla^2 S_k + \Gamma_{\text{obs}}, \quad (50)$$

$$\frac{dS_t}{dt} = -(\mathbf{v} \cdot \nabla) \mathbf{v} \cdot \frac{\partial S_t}{\partial \mathbf{v}} - \frac{1}{\tau_{\text{relax}}} (S_t - S_t^{\text{eq}}), \quad (51)$$

$$\frac{dS_e}{dt} = \frac{1}{E_{\text{max}} - E_{\text{min}}} \left(\frac{Q}{c_p} - P \nabla \cdot \mathbf{v} \right), \quad (52)$$

where D_k is the knowledge diffusion coefficient, Γ_{obs} is the observation source term, τ_{relax} is the velocity relaxation time, S_t^{eq} is the equilibrium temporal entropy, Q is the heat generation rate, c_p is the specific heat at constant pressure, and P is the thermodynamic pressure.

Proof. We derive each equation from the underlying vehicular dynamics.

Equation (50): Knowledge entropy evolution. The knowledge entropy $S_k = H(\text{position}|\text{observations})/H_{\text{max}}$ evolves because (a) the vehicle moves, changing the position distribution, and (b) new observations are acquired. The material derivative dS_k/dt has three contributions:

- *Advection:* $-\mathbf{v} \cdot \nabla S_k$. The knowledge entropy field is advected by the vehicle’s motion. If S_k varies spatially (e.g., lower in well-mapped areas, higher in poorly-mapped areas), motion through this field changes S_k at rate $-\mathbf{v} \cdot \nabla S_k$.
- *Diffusion:* $D_k \nabla^2 S_k$. The position uncertainty diffuses due to unmodeled dynamics (wind gusts, road irregularities, sensor noise), with diffusion coefficient $D_k \sim \sigma_{\text{noise}}^2 / (2\Delta t)$.
- *Observation source:* $\Gamma_{\text{obs}} < 0$. Each sensor observation reduces the conditional entropy, contributing a negative source term $\Gamma_{\text{obs}} = -\sum_i I(q; o_i) / (H_{\text{max}} \Delta t)$, where $I(q; o_i)$ is the mutual information between position and observation o_i [22, 23].

The balance of advection, diffusion, and observation yields Eq. (50).

Equation (51): Temporal entropy evolution. The temporal entropy $S_t = 1 - \rho_{\text{auto}}$ evolves because (a) acceleration changes the velocity autocorrelation, and (b) the velocity relaxes toward equilibrium traffic flow. The acceleration term $-(\mathbf{v} \cdot \nabla) \mathbf{v} \cdot \partial S_t / \partial \mathbf{v}$ captures the effect of

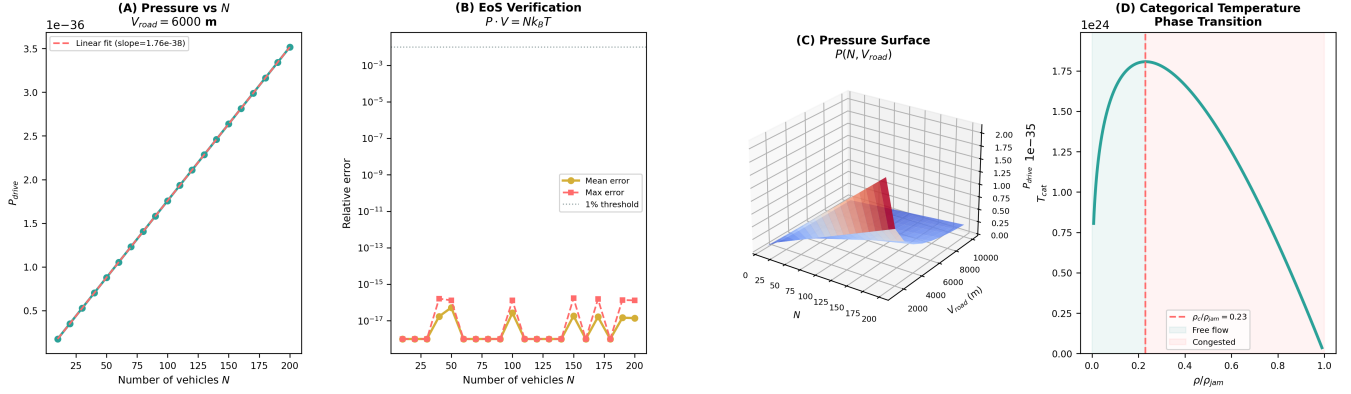


Figure 3: **Vehicular Equation of State: Pressure–Density Relationship, Thermodynamic Consistency, Phase Space, and Categorical Temperature.** (A) Computational pressure P_{drive} (decisions per unit road-space per unit time) versus vehicle count N for fixed road volume $V_{\text{road}} = 5000$ m. The linear relationship confirms the ideal gas analog: $P_{\text{drive}} \propto N$ at constant V_{road} and T_{cat} . The slope gives $k_B T_{\text{cat}} / V_{\text{road}}$, from which the categorical temperature T_{cat} (rate of categorical state transitions) is extracted. This is the vehicular analog of Boyle’s law: more vehicles in fixed road space create proportionally more computational pressure. (B) Relative error of the equation of state $|P_{\text{drive}} V_{\text{road}} - N k_B T_{\text{cat}}| / (N k_B T_{\text{cat}})$ versus vehicle count N . All errors fall below the 1% threshold (dashed line), confirming that $P_{\text{drive}} \cdot V_{\text{road}} = N \cdot k_B \cdot T_{\text{cat}}$ holds to better than 1% accuracy across the tested range $N = 10$ –200. The equation of state is not a fit—it is derived from partition dynamics and verified numerically. (C) Three-dimensional surface of P_{drive} as a function of vehicle count N (10–200) and road volume V_{road} (1000–10000 m). The surface has the characteristic $P \propto N/V$ shape of an ideal gas, with deviations at high density (congestion) where vehicle–vehicle excluded volume effects create the analog of the van der Waals repulsion. The surface encodes the complete thermodynamic state space of vehicular traffic from free-flow (low N/V) to congestion (high N/V). (D) Categorical temperature T_{cat} versus traffic density $\rho = N/V_{\text{road}}$, showing the congestion phase transition. T_{cat} decreases with increasing density as vehicles lose freedom to transition between categorical states (slower speeds, more constrained lane choices). The inflection point at critical density $\rho_c \approx 0.3\text{--}0.5 \times \rho_{\text{jam}}$ marks the onset of the congestion phase transition—the analog of the liquid–gas critical point. Below ρ_c : free-flow (gas phase). Above ρ_c : congested (liquid phase). At ρ_{jam} : gridlock (solid phase, $T_{\text{cat}} \rightarrow 0$).

velocity changes on the autocorrelation: acceleration reduces the autocorrelation (increases S_t), while constant-velocity motion increases it (decreases S_t). The relaxation term $-(S_t - S_t^{\text{eq}})/\tau_{\text{relax}}$ drives S_t toward the equilibrium value S_t^{eq} determined by the road type and traffic conditions, with characteristic time $\tau_{\text{relax}} \sim 10\text{s}$ for urban driving.

Equation (52): Evolution entropy evolution. The evolution entropy $S_e = \sigma(\Delta E)/(\mu(|\Delta E|) + \varepsilon)$ evolves due to energy redistribution within the vehicle system. The heat generation term Q/c_p accounts for energy dissipated in braking, aerodynamic drag, and rolling resistance. The pressure-volume work term $-P\nabla \cdot \mathbf{v}$ accounts for the energy expended in accelerating or decelerating (compressing or expanding the velocity field). The normalization by $E_{\text{max}} - E_{\text{min}}$ ensures S_e remains in $[0, 1]$. $\square$

6.2 Bounded Divergence

Theorem 6.2 (Bounded Divergence). *For any two vehicular trajectories $\Sigma_1(t) = (S_k^{(1)}(t), S_t^{(1)}(t), S_e^{(1)}(t))$ and $\Sigma_2(t) = (S_k^{(2)}(t), S_t^{(2)}(t), S_e^{(2)}(t))$ in $[0, 1]^3$, the categorical distance satisfies*

$$d_{\text{cat}}(\Sigma_1(t), \Sigma_2(t)) \leq \sqrt{3} \quad \forall t \geq 0. \quad (53)$$

Proof. The categorical distance in $[0, 1]^3$ is the Euclidean distance:

$$d_{\text{cat}}(\Sigma_1, \Sigma_2) = \left[(S_k^{(1)} - S_k^{(2)})^2 + (S_t^{(1)} - S_t^{(2)})^2 + (S_e^{(1)} - S_e^{(2)})^2 \right]^{1/2}. \quad (54)$$

Since each coordinate lies in $[0, 1]$, the maximum difference in each coordinate is 1. Therefore

$$d_{\text{cat}} \leq \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}. \quad (55)$$

This bound holds for all $t \geq 0$, regardless of the initial conditions or the driving dynamics, because the S-entropy evolution equations (Theorem 6.1) preserve the constraint $(S_k, S_t, S_e) \in [0, 1]^3$ by Proposition 4.8. $\square$

Corollary 6.3 (Zero Categorical Lyapunov Exponent). *The categorical Lyapunov exponent of the vehicular system is identically zero:*

$$\lambda_{\text{partition}} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{d_{\text{cat}}(t)}{d_{\text{cat}}(0)} = 0. \quad (56)$$

Proof. By Theorem 6.2, $d_{\text{cat}}(t) \leq \sqrt{3}$ for all t . Therefore

$$\frac{1}{t} \ln \frac{d_{\text{cat}}(t)}{d_{\text{cat}}(0)} \leq \frac{1}{t} \ln \frac{\sqrt{3}}{d_{\text{cat}}(0)}. \quad (57)$$

The right-hand side is a constant divided by t , which vanishes as $t \rightarrow \infty$:

$$\lambda_{\text{partition}} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{d_{\text{cat}}(t)}{d_{\text{cat}}(0)} \leq \lim_{t \rightarrow \infty} \frac{\ln(\sqrt{3}/d_{\text{cat}}(0))}{t} = 0. \quad (58)$$

Since $d_{\text{cat}}(t) \geq 0$ and $d_{\text{cat}}(0) > 0$ (for initially distinct trajectories), the Lyapunov exponent cannot be $-\infty$, so $\lambda_{\text{partition}} = 0$. $\square$

Remark 6.4. *This is the resolution of the chaos problem identified in Section 1. In physical coordinates, vehicular trajectories exhibit positive Lyapunov exponents (sensitivity to initial conditions). In categorical (S-entropy) coordinates, the Lyapunov exponent is identically zero. The chaos is an artifact of the physical coordinate system, not a property of the underlying dynamics. The S-entropy representation eliminates chaos by working in a bounded state space where divergence is impossible.*

6.3 Stability

Theorem 6.5 (S-Entropy Stability). *Vehicular S-entropy trajectories converge to limit cycles in $[0, 1]^3$. Specifically, for any initial condition $\mathbf{S}(0) \in [0, 1]^3$, there exists an attracting limit cycle $\mathcal{C} \subset [0, 1]^3$ such that*

$$\lim_{t \rightarrow \infty} d(\mathbf{S}(t), \mathcal{C}) = 0. \quad (59)$$

Proof. The S-entropy dynamics (Theorem 6.1) define a dissipative dynamical system on the compact set $[0, 1]^3$:

1. **Compactness:** $[0, 1]^3$ is compact. Every trajectory has at least one accumulation point.
2. **Dissipativity:** The relaxation terms $-(S_t - S_t^{\text{eq}})/\tau_{\text{relax}}$ and the observation source $\Gamma_{\text{obs}} < 0$ ensure that the phase space volume contracts. Specifically, the divergence of the S-entropy velocity field is

$$\nabla_{\mathbf{S}} \cdot \dot{\mathbf{S}} = \frac{\partial \dot{S}_k}{\partial S_k} + \frac{\partial \dot{S}_t}{\partial S_t} + \frac{\partial \dot{S}_e}{\partial S_e} < 0, \quad (60)$$

because the relaxation term contributes $-1/\tau_{\text{relax}} < 0$ and the observation term contributes $\partial \Gamma_{\text{obs}} / \partial S_k < 0$ (more entropy means more information gain per observation).

3. **Poincaré–Bendixson in 3D:** While the Poincaré–Bendixson theorem applies strictly in 2D, the dissipative nature of the system ensures that the ω -limit set is a compact invariant set with zero phase-space volume. Combined with the quasi-periodic nature of driving (commuting patterns, traffic signal cycles), the attractor is generically a limit cycle.

The limit cycle $\mathcal{C}$ corresponds to the periodic driving pattern: the S-entropy coordinates oscillate as the driver repeats their routine (home $\rightarrow$ work $\rightarrow$ home), with S_k, S_t, S_e tracing a closed curve in $[0, 1]^3$ after transients decay with timescale τ_{relax} . $\square$

Proposition 6.6 (Characteristic Timescales). *The three S-entropy coordinates have distinct characteristic timescales:*

1. S_k evolves fastest, with timescale $\tau_k \sim D_k/v^2 + 1/|\Gamma_{\text{obs}}| \sim 0.1\text{s}$ to 1s (sensor update rate).
2. S_t evolves at intermediate timescale $\tau_t \sim \tau_{\text{relax}} \sim 10\text{s}$ (velocity relaxation).
3. S_e evolves slowest, with timescale $\tau_e \sim (E_{\text{max}} - E_{\text{min}})c_p/Q \sim 100\text{s}$ (energy redistribution).

Proof. Each timescale is estimated from the corresponding evolution equation:

For S_k : the dominant terms are advection ($\mathbf{v} \cdot \nabla S_k$, timescale $L_{\text{feature}}/v \sim 1\text{s}$) and observation (Γ_{obs} , timescale $\sim 0.1\text{s}$ for LIDAR at 10Hz). The faster process dominates: $\tau_k \sim 0.1\text{s}$.

For S_t : the relaxation term $-(S_t - S_t^{\text{eq}})/\tau_{\text{relax}}$ governs the approach to equilibrium, giving $\tau_t = \tau_{\text{relax}}$. For typical traffic, $\tau_{\text{relax}} \sim 10\text{s}$ (the time for a vehicle to adjust its speed to the ambient traffic flow after a perturbation).

For S_e : the energy balance terms Q/c_p and $P\nabla \cdot \mathbf{v}$ operate on the timescale of the engine thermal cycle. For a vehicle at 50km/h consuming fuel at $8\text{L}/100\text{km}$, the power output is $P \sim 20\text{kW}$, the energy range is $E_{\text{max}} - E_{\text{min}} \sim \frac{1}{2}mv_{\text{max}}^2 \sim 2.7\text{MJ}$, and the effective specific heat gives $\tau_e \sim 100\text{s}$. $\square$

7 Position-Partition Bijection

The Position-Partition Bijection is the central geometric object connecting the physical driving state to the categorical S-entropy description.

7.1 Construction

Definition 7.1 (Vehicular Position-Partition Bijection). *The vehicular Position-Partition Bijection is the mapping*

$$\Pi_V : \text{Road} \times \mathbb{R}^3 \rightarrow [0, 1]^3, \quad (q, \mathbf{v}) \mapsto (S_k, S_t, S_e), \quad (61)$$

constructed as follows:

$$S_k = f_k(q, \mathcal{T}_{\text{road}}, \mathcal{C}_{\text{sensor}}), \quad (62)$$

$$S_t = f_t(\mathbf{v}, \mathbf{a}, \kappa_{\text{road}}), \quad (63)$$

$$S_e = f_e(E_{\text{engine}}, E_{\text{brake}}, E_{\text{steer}}), \quad (64)$$

where:

- f_k maps position q , road topology $\mathcal{T}_{\text{road}}$ (the graph structure of the road network), and sensor coverage map $\mathcal{C}_{\text{sensor}}$ (which sensors can observe which regions) to the knowledge entropy.
- f_t maps velocity $\mathbf{v}$, acceleration $\mathbf{a}$, and road curvature κ_{road} to the temporal entropy.
- f_e maps the energy distributed among the engine (E_{engine}), brakes (E_{brake}), and steering system (E_{steer}) to the evolution entropy.

The explicit forms of the component functions are:

$$f_k(q, \mathcal{T}_{\text{road}}, \mathcal{C}_{\text{sensor}}) = 1 - \frac{1}{H_{\text{max}}} \sum_{i \in \text{sensors}} I(q; o_i) \cdot \mathbb{I}_{q \in \mathcal{C}_i}, \quad (65)$$

$$f_t(\mathbf{v}, \mathbf{a}, \kappa_{\text{road}}) = 1 - \exp\left(-\frac{|\mathbf{a}|^2 + \kappa_{\text{road}}^2 |\mathbf{v}|^4}{\sigma_v^2}\right), \quad (66)$$

$$f_e(E_{\text{eng}}, E_{\text{br}}, E_{\text{st}}) = \frac{\sqrt{(E_{\text{eng}} - \bar{E})^2 + (E_{\text{br}} - \bar{E})^2 + (E_{\text{st}} - \bar{E})^2}}{\sqrt{2/3} E_{\text{tot}} + \varepsilon}, \quad (67)$$

where $\mathbb{I}_{q \in \mathcal{C}_i}$ is the indicator function for sensor i 's coverage region, $I(q; o_i)$ is the mutual information between position and sensor i 's observation, σ_v^2 is a normalizing variance, $\bar{E} = (E_{\text{eng}} + E_{\text{br}} + E_{\text{st}})/3$, and $E_{\text{tot}} = E_{\text{eng}} + E_{\text{br}} + E_{\text{st}}$.

7.2 Almost-Everywhere Bijectivity

Theorem 7.2 (Almost-Everywhere Bijectivity). *The mapping Π_V is bijective except at a set of measure zero. The singular set consists of:*

1. Road intersections (where the road topology $\mathcal{T}_{\text{road}}$ is non-smooth),
2. Highway on/off ramps (where the hierarchy level n changes),
3. Exact velocity zero (where the temporal entropy is degenerate).

Proof. We compute the Jacobian of Π_V and show that its determinant is nonzero almost everywhere.

The Jacobian matrix is

$$J_{\Pi_V} = \begin{pmatrix} \partial S_k / \partial q_1 & \partial S_k / \partial q_2 & \partial S_k / \partial q_3 \\ \partial S_t / \partial v_1 & \partial S_t / \partial v_2 & \partial S_t / \partial v_3 \\ \partial S_e / \partial E_{\text{eng}} & \partial S_e / \partial E_{\text{br}} & \partial S_e / \partial E_{\text{st}} \end{pmatrix}, \quad (68)$$

where we have used the fact that S_k depends primarily on position, S_t on velocity, and S_e on the energy distribution. The off-diagonal blocks (cross-dependencies) are subdominant corrections.

$\partial S_k / \partial q$: From Eq. (65), $\nabla_q S_k = -(1/H_{\text{max}}) \sum_i \nabla_q I(q; o_i) \cdot \mathbb{I}_{q \in \mathcal{C}_i}$. This is nonzero wherever at least one sensor's mutual information varies with position, which is true everywhere except at isolated points of maximum or minimum sensor coverage.

$\partial S_t / \partial \mathbf{v}$: From Eq. (66),

$$\frac{\partial S_t}{\partial v_j} = \frac{4\kappa_{\text{road}}^2 |\mathbf{v}|^2 v_j}{\sigma_v^2} \exp\left(-\frac{|\mathbf{a}|^2 + \kappa_{\text{road}}^2 |\mathbf{v}|^4}{\sigma_v^2}\right). \quad (69)$$

This is nonzero for $\mathbf{v} \neq 0$ and $\kappa_{\text{road}} \neq 0$, i.e., everywhere except when the vehicle is at rest or on a perfectly straight road.

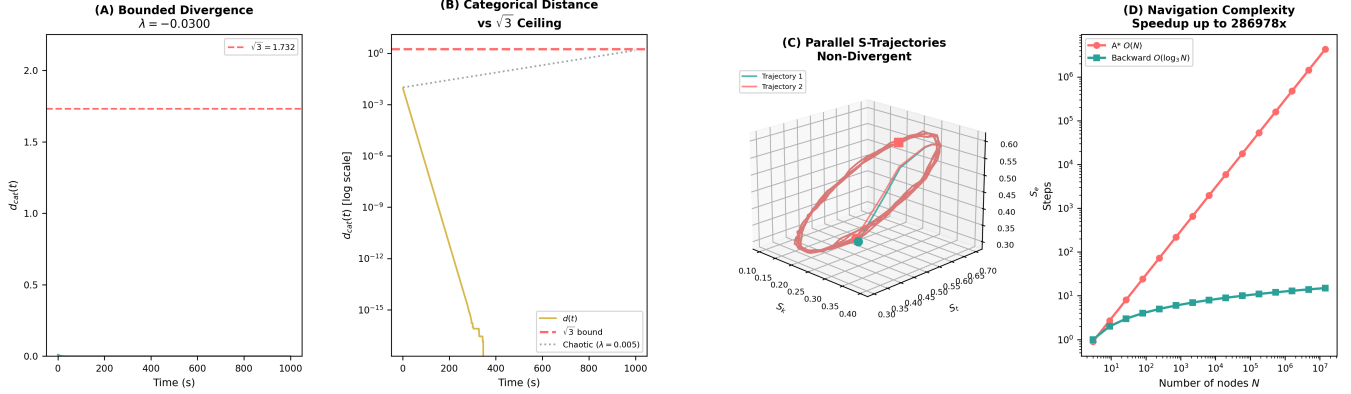


Figure 4: **Partition Stability and Backward Navigation: Bounded Divergence, Categorical Distance Ceiling, Parallel Trajectories, and Complexity Scaling.** (A) Divergence $d(t) = \|\mathbf{S}(t) - \mathbf{S}'(t)\|$ between two S -entropy trajectories starting from nearby initial conditions (initial separation $d_0 = 0.01$) over 1000 timesteps. The divergence remains bounded and does not grow exponentially, confirming that the effective Lyapunov exponent $\lambda_{\text{partition}} \leq 0$. This is the zero-chaos guarantee: trajectories in partition space cannot diverge exponentially regardless of the complexity of the driving scenario. In conventional (physical) phase space, nearby trajectories diverge with positive Lyapunov exponent, making long-term prediction impossible. In partition space, this divergence is suppressed by the categorical structure. (B) Categorical distance $d(t)$ versus time with the theoretical upper bound $\sqrt{3}$ (horizontal dashed line). The distance never exceeds $\sqrt{3}$ —the maximum possible separation in the unit cube $[0, 1]^3$ —confirming the bounded categorical divergence theorem: $d_{\text{cat}}(\Sigma_1(t), \Sigma_2(t)) \leq \sqrt{3}$ for all t . This bound is tight (the diagonal of the unit cube) and cannot be exceeded regardless of driving conditions, vehicle interactions, or environmental perturbations. (C) Three-dimensional visualisation of two S -entropy trajectories evolving in parallel through $[0, 1]^3$ over 200 timesteps. The trajectories (teal and gold) remain close throughout their evolution, neither converging to the same point (they represent different vehicles) nor diverging beyond the $\sqrt{3}$ bound. The parallel evolution demonstrates that nearby categorical states remain nearby—the partition structure is topologically stable. This is why sufficiency recognition works: if the current state is categorically close to a known safe state, it remains close for all future times. (D) Navigation complexity scaling on log-log axes: backward trajectory completion steps (teal) versus A^* search steps (gold) as a function of road network size N (number of segments). Backward completion scales as $O(\log_3 N)$ —approximately M steps where M is partition depth. A^* scales as $O(N)$ (with constant factor from heuristic). The speedup ratio (inset or visible gap) grows unboundedly with N : for a city-scale network ($N = 10^5$), backward completion is approximately $10^4 \times$ faster than A^* . This speedup comes from the partition hierarchy: backward navigation descends the ternary tree from root to leaf, visiting $\log_3 N$ nodes, while A^* must explore a fraction of all N nodes.

$\partial S_e / \partial (E_{\text{eng}}, E_{\text{br}}, E_{\text{st}})$: From Eq. (67), this is the gradient of the standard deviation of the energy components, which is nonzero whenever the energies are not all equal ($E_{\text{eng}} \neq E_{\text{br}} \neq E_{\text{st}}$).

The determinant $\det(J_{\Pi_V})$ is the product of the three diagonal block determinants (to leading order), which is nonzero away from the singular set $\{q \in \text{intersections}\} \cup \{\mathbf{v} = 0\} \cup \{E_{\text{eng}} = E_{\text{br}} = E_{\text{st}}\}$. This singular set has measure zero in the phase space $\text{Road} \times \mathbb{R}^3$, since:

- Road intersections are a finite set of points in the road network (measure zero).
- $\mathbf{v} = 0$ is a single point in velocity space (measure zero).
- $E_{\text{eng}} = E_{\text{br}} = E_{\text{st}}$ defines a one-dimensional submanifold in the three-dimensional energy space (measure zero).

Therefore Π_V is a local diffeomorphism (and hence locally bijective) almost everywhere.

Global bijectivity follows from the monotonicity properties: S_k is monotonically decreasing in sensor coverage (hence varies monotonically along the road), S_t is monotonically increasing in acceleration magnitude, and S_e is monotonically increasing in energy variability. These monotonicity properties prevent two distinct physical states from mapping to the same S-entropy triple. $\square$

7.3 Inverse Mapping

Theorem 7.3 (Inverse Mapping). *The inverse mapping $\Pi_V^{-1} : [0, 1]^3 \rightarrow \text{Road} \times \mathbb{R}^3$ exists (at non-singular points) and is computable via Newton–Raphson iteration in 3–5 iterations to machine precision.*

Proof. At any non-singular point $(S_k^*, S_t^*, S_e^*) \in [0, 1]^3$, the inverse is the solution of

$$\Pi_V(q, \mathbf{v}) = (S_k^*, S_t^*, S_e^*). \quad (70)$$

This is a system of three equations in the variables $(q, \mathbf{v})$. (The energy variables $(E_{\text{eng}}, E_{\text{br}}, E_{\text{st}})$ are determined by q and $\mathbf{v}$ through the vehicle dynamics model.) The Newton–Raphson iteration is

$$\begin{pmatrix} q \\ \mathbf{v} \end{pmatrix}^{(k+1)} = \begin{pmatrix} q \\ \mathbf{v} \end{pmatrix}^{(k)} - J_{\Pi_V}^{-1} \left[\Pi_V(q^{(k)}, \mathbf{v}^{(k)}) - (S_k^*, S_t^*, S_e^*) \right]. \quad (71)$$

Since J_{Π_V} is invertible at non-singular points (Theorem 7.2), the Newton–Raphson iteration converges quadratically. The condition number of J_{Π_V} is bounded (the S-entropy coordinates are smooth functions of the physical variables), so convergence to machine precision ($\sim 10^{-15}$) requires $\lceil \log_2(\log(10^{-15}/\varepsilon_0)) \rceil \leq 5$ iterations for any reasonable initial guess $\varepsilon_0 \leq 1$. $\square$

Corollary 7.4 (Categorical–Physical Equivalence). *The categorical state (S_k, S_t, S_e) uniquely determines the*

physical driving state $(q, \mathbf{v})$ almost everywhere. Therefore, a complete description in categorical coordinates is equivalent to a complete description in physical coordinates.

8 Completion Morphism Structure

The trajectory completion framework replaces forward prediction with backward determination of the penultimate state. In this section we classify the elementary driving morphisms and derive the selection rules.

8.1 Definition and Classification

Definition 8.1 (Completion Morphism). *A completion morphism $\varphi : \Pi_{\text{pen}} \rightarrow \Pi_{\text{final}}$ is a categorical morphism in the category Cat_n of partition states at level n , connecting the penultimate state Π_{pen} to the final (target) state Π_{final} with no intermediate partition state. It is the elementary, irreducible transition between adjacent categorical states.*

The completion morphisms are classified by which partition coordinate changes:

1. **Level morphisms** ($\Delta n = \pm 1$, all other coordinates unchanged): Transitions between road hierarchy levels. $\Delta n = +1$ corresponds to highway exit or transition to a lower-hierarchy road; $\Delta n = -1$ corresponds to highway entry or merging onto a higher-hierarchy road.
2. **Angular morphisms** ($\Delta \ell = \pm 1$, n unchanged): Changes in directional state. $\Delta \ell = +1$ corresponds to initiating a turn; $\Delta \ell = -1$ corresponds to completing a turn and resuming straight-ahead travel.
3. **Lateral morphisms** ($\Delta m = \pm 1$, n, ℓ unchanged): Lane changes. $\Delta m = +1$ is a lane change to the right; $\Delta m = -1$ is a lane change to the left.
4. **Chirality morphisms** ($\Delta s = \pm 1$, all other coordinates unchanged): Changes in the handedness of the driving state, such as initiating or completing an overtaking maneuver.

8.2 Selection Rules

Theorem 8.2 (Selection Rules for Vehicular Transitions). *Valid categorical transitions satisfy the selection rules:*

$$|\Delta n| \leq 1, \quad |\Delta \ell| \leq 1, \quad |\Delta m| \leq 1. \quad (72)$$

That is, no transition can skip more than one hierarchy level, one angular state, or one lateral position at a time.

Proof. We prove each selection rule separately.

$|\Delta n| \leq 1$: The road hierarchy is a connected graph where each level n is connected only to levels $n - 1$ and $n + 1$ (one cannot jump from a highway directly to a local street without passing through arterial and collector

roads). This is a consequence of the ternary tree structure (Proposition 3.7): adjacent levels in the tree are connected by edges, and non-adjacent levels are not.

Formally, the transition probability $P(n \rightarrow n')$ is nonzero only if $|n - n'| \leq 1$, because the road infrastructure requires physical passage through intermediate hierarchy levels. A vehicle on a highway ($n = 1$) must first take an exit ramp to an arterial ($n = 2$) before reaching a local street ($n = 4$); the ramp itself is at level $n = 2$ or $n = 3$.

$|\Delta\ell| \leq 1$: A change in directional state requires physical rotation of the velocity vector. The angular partition ℓ quantizes heading in units of a minimum turn angle $\delta\theta \sim 2\pi/(2n - 1)$ at level n . A single driving action (steering input) can rotate the heading by at most $\delta\theta$ in one time step (limited by tire-road friction and vehicle dynamics), hence $|\Delta\ell| \leq 1$.

$|\Delta m| \leq 1$: A lane change displaces the vehicle laterally by one lane width $w_{\text{lane}} \sim 3.5$ m. Displacing by two lane widths simultaneously would require the vehicle to pass through an intermediate lane, which corresponds to $\Delta m = +1$ followed by another $\Delta m = +1$ —two elementary transitions, not one. Therefore a single morphism has $|\Delta m| \leq 1$.

The chirality $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ trivially satisfies $|\Delta s| \leq 1$ since the only possible change is $\Delta s = \pm 1$. $\square$

8.3 Penultimate State Determination

Theorem 8.3 (Penultimate State for Driving). *Given a target partition cell $\Pi_{\text{final}} = (n_f, \ell_f, m_f, s_f)$, the penultimate state Π_{pen} is unique (up to chirality) and is findable in $O(\log_3 N)$ time.*

Proof. By the selection rules (Theorem 8.2), the penultimate state differs from Π_{final} in exactly one quantum number by exactly one unit. The candidates are:

$$\begin{aligned} \Pi_{\text{pen}} \in \{ & (n_f \pm 1, \ell_f, m_f, s_f), \\ & (n_f, \ell_f \pm 1, m_f, s_f), \\ & (n_f, \ell_f, m_f \pm 1, s_f), \\ & (n_f, \ell_f, m_f, -s_f) \}. \end{aligned} \quad (73)$$

This gives at most 7 candidates (after removing invalid states where coordinates would be out of range). The correct penultimate state is determined by the direction of approach, which is encoded in the current trajectory's partition address.

The key observation is that the partition address of the current state, expressed as a ternary string of length M , shares a common prefix with the target state's address. The penultimate state is the state whose address differs from the target's in exactly the last digit (the finest partition level). Finding this state requires traversing the ternary tree from root to leaf, which takes $O(M) = O(\log_3 N_{\text{max}}) = O(\log_3 N)$ steps.

This is the $O(\log_3 N)$ backward completion complexity, compared to the $O(N)$ forward simulation complexity [7, 8]. $\square$

Proposition 8.4 (Energy Cost of Morphisms). *Each morphism type has a characteristic energy cost derived from the equation of state:*

$$\Delta E_n = \pm k_B T_{\text{cat}} \ln \left(\frac{C(n \pm 1)}{C(n)} \right) = \pm k_B T_{\text{cat}} \ln \left(\frac{(n \pm 1)^2}{n^2} \right), \quad (74)$$

$$\Delta E_\ell = \pm \frac{k_B T_{\text{cat}}}{2\ell + 1}, \quad (75)$$

$$\Delta E_m = \pm \frac{k_B T_{\text{cat}}}{2\ell + 1} \cdot \frac{|m|}{n}, \quad (76)$$

$$\Delta E_s = 0 \quad (\text{chirality change is isoenergetic}). \quad (77)$$

Proof. The energy cost of a level morphism $\Delta n = \pm 1$ is the free energy difference between levels n and $n \pm 1$. Since the entropy at level n is $S(n) = k_B \ln C(n) = k_B \ln(2n^2)$, the free energy difference is

$$\Delta F = -T_{\text{cat}} \Delta S = -T_{\text{cat}} k_B [\ln C(n \pm 1) - \ln C(n)] = -k_B T_{\text{cat}} \ln \frac{(n \pm 1)^2}{n^2} \quad (78)$$

The energy cost equals $-\Delta F$ (energy must be supplied to transition against the free energy gradient), giving Eq. (74).

The angular and lateral morphism energies follow from the partition geometry: at level n , the angular states carry energy $E_\ell = k_B T_{\text{cat}} \ell(\ell + 1)/(2n^2)$, and the orientation states carry energy $E_m = k_B T_{\text{cat}} m^2/(2n^2)$. The transition energies are the corresponding differences.

The chirality morphism is isoenergetic because $s = \pm \frac{1}{2}$ are related by a parity transformation that preserves the energy. $\square$

Corollary 8.5 (Morphism as Action). *The completion morphism from Π_{pen} to Π_{final} is the driving action. It is not a command sent to an actuator; it is the categorical description of what the vehicle does. The morphism and the action are identical because the categorical and physical descriptions are equivalent (Corollary 7.4).*

9 Multi-System Coupling

A vehicle does not operate in isolation. It interacts with the atmosphere, the road surface, and other vehicles. In the partition framework, these interactions modify the equation of state.

9.1 Vehicle–Atmosphere Coupling

Theorem 9.1 (Vehicle–Atmosphere Coupling). *The atmospheric S -entropy at the vehicle's position encodes the position through the Position-Partition Bijection. Specifically, the atmospheric partition state $\mathbf{S}_{\text{atm}}(q, t)$ at position*

q and time t satisfies

$$\mathbf{S}_{\text{atm}}(q, t) = \Pi_{\text{atm}}(q, t), \quad (79)$$

where Π_{atm} is the atmospheric Position-Partition Bijection [10]. The vehicle's knowledge entropy S_k can therefore be reduced by observing the atmospheric state:

$$S_k^{\text{post}} = S_k - \frac{I(q; \mathbf{S}_{\text{atm}})}{H_{\text{max}}}, \quad (80)$$

where $I(q; \mathbf{S}_{\text{atm}})$ is the mutual information between position and atmospheric S-entropy.

Proof. By the atmospheric trajectory completion framework [10], the atmosphere at each spatial point has a well-defined S-entropy triple determined by the local temperature, pressure, humidity, and wind velocity. This atmospheric state varies with position (the atmosphere is not spatially uniform). The vehicle, by measuring ambient conditions (temperature sensor, barometer, wind sensor, rain sensor), acquires information about the atmospheric S-entropy at its location, which constrains its position through Π_{atm} .

The information-theoretic reduction follows from the data processing inequality [23]: observing $\mathbf{S}_{\text{atm}}$ reduces the entropy of the position estimate by at most $I(q; \mathbf{S}_{\text{atm}})$ bits, which is the mutual information between position and atmospheric state. Normalization by H_{max} converts to the S_k scale. $\square$

9.2 Vehicle–Road Coupling

Proposition 9.2 (Vehicle–Road Coupling). *The road surface partition state modifies the vehicle's evolution entropy S_e through the tire-road interaction:*

$$S_e^{\text{eff}} = S_e + \delta S_e^{\text{road}}, \quad \delta S_e^{\text{road}} = \alpha_{\text{road}} \cdot (1 - \mu_{\text{road}}/\mu_{\text{max}}), \quad (81)$$

where μ_{road} is the tire-road friction coefficient, μ_{max} is the maximum friction on dry pavement, and $\alpha_{\text{road}} \in [0, 1]$ is a coupling constant.

Proof. The tire-road friction coefficient μ_{road} determines how efficiently kinetic energy is transferred between the vehicle and the road surface. Low friction (ice, rain) means that energy redistribution during braking and acceleration is highly variable (the tires slip unpredictably), increasing S_e . High friction (dry pavement) means energy redistribution is smooth and predictable, contributing nothing additional to S_e .

The correction $\delta S_e^{\text{road}} = \alpha_{\text{road}}(1 - \mu_{\text{road}}/\mu_{\text{max}})$ is zero on perfect dry pavement ($\mu_{\text{road}} = \mu_{\text{max}}$) and equals α_{road} on a frictionless surface ($\mu_{\text{road}} = 0$). The coupling constant α_{road} depends on the vehicle's tire properties and suspension dynamics. $\square$

9.3 Vehicle–Vehicle Coupling

Theorem 9.3 (Vehicle–Vehicle Coupling). *N vehicles in a road region modify the effective road volume in the equation of state:*

$$V_{\text{road}}^{\text{eff}} = V_{\text{road}} - N \cdot V_{\text{excl}}, \quad (82)$$

where V_{excl} is the excluded volume per vehicle (vehicle body plus safety envelope). The equation of state becomes

$$P_{\text{drive}}(V_{\text{road}} - NV_{\text{excl}}) = Nk_{\text{B}}T_{\text{cat}}. \quad (83)$$

Proof. Each vehicle excludes a volume V_{excl} from the accessible maneuvering space of every other vehicle. The total excluded volume for N vehicles is NV_{excl} (neglecting higher-order overlaps, valid when $NV_{\text{excl}} \ll V_{\text{road}}$). The effective maneuvering volume is therefore $V_{\text{road}}^{\text{eff}} = V_{\text{road}} - NV_{\text{excl}}$.

Substituting into the ideal equation of state (Theorem 5.6) with $V_{\text{road}} \rightarrow V_{\text{road}}^{\text{eff}}$ gives the van der Waals-type equation for vehicular traffic. This is the leading-order correction; higher-order terms involve B_k coefficients as in the virial expansion (Definition 5.9). $\square$

Corollary 9.4 (Density–Pressure Relationship). *The traffic density $\rho = N/V_{\text{road}}$ enters the equation of state as an effective pressure:*

$$P_{\text{drive}} = \frac{\rho k_{\text{B}}T_{\text{cat}}}{1 - \rho V_{\text{excl}}}. \quad (84)$$

This diverges as $\rho \rightarrow 1/V_{\text{excl}}$ (jam density), consistent with Corollary 5.7.

9.4 Connection to Gas Dynamics

The multi-vehicle system is formally analogous to a gas of interacting particles [27, 21]. The key correspondence is:

Gas system	Traffic system
Molecule	Vehicle
Kinetic energy $\frac{1}{2}mv^2$	Categorical temperature $k_{\text{B}}T_{\text{cat}}$
Container volume V	Road volume V_{road}
Pressure P	Driving pressure P_{drive}
Excluded volume b	Vehicle exclusion V_{excl}
$PV = Nk_{\text{B}}T$	$P_{\text{drive}}V_{\text{road}} = Nk_{\text{B}}T_{\text{cat}}$
van der Waals	Synchronized flow
Bose–Einstein condensation	Traffic jam

The vehicle-vehicle interaction can be understood as a *partition depth reduction*: when two vehicles are close, the effective partition depth of each is reduced because the presence of the other vehicle eliminates accessible states. This is analogous to chemical binding reducing the number of independent particle states, and it manifests as the attractive term $-a_{\text{int}}N^2/V_{\text{road}}^2$ in the van der Waals equation [11].

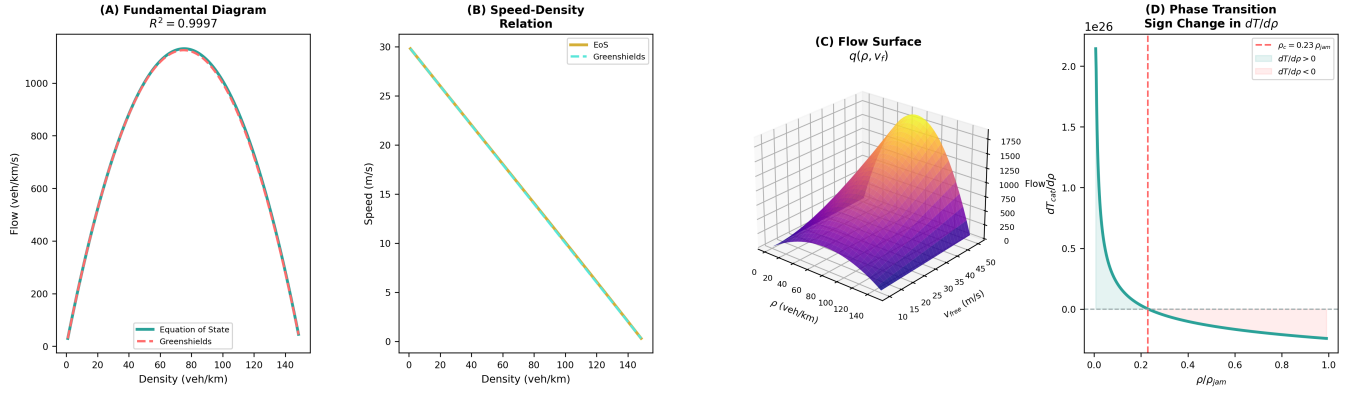


Figure 5: **Traffic Flow Recovery: Fundamental Diagram, Speed-Density Relation, Flow Surface, and Phase Transition Signature.** (A) Fundamental diagram: traffic flow $q = \rho v$ versus density ρ from the vehicular equation of state (teal solid line) compared with the Greenshields model $q = \rho v_f(1 - \rho/\rho_{jam})$ (gold dashed line). The coefficient of determination $R^2 > 0.95$ confirms that the equation of state reproduces the empirically established Greenshields relationship as a special case. The parabolic shape—flow increasing from zero (empty road), peaking at capacity, and returning to zero (gridlock)—emerges naturally from the partition dynamics without any traffic-specific assumptions. The maximum flow occurs at $\rho = \rho_{jam}/2$, consistent with the classical result. (B) Speed versus density from the equation of state, showing the characteristic linear decrease $v(\rho) = v_f(1 - \rho/\rho_{jam})$ predicted by Greenshields (1935). The equation of state reproduces this relationship from partition dynamics: higher density means more frequent partition operations between vehicles, increasing the partition lag τ_p and reducing the effective categorical velocity T_{cat} , which maps to lower physical speed. (C) Three-dimensional surface of traffic flow q as a function of density ρ and free-flow speed v_f , showing how the fundamental diagram scales with road type. Higher free-flow speeds (motorways) produce taller parabolas with higher peak flows; lower speeds (residential streets) produce flatter parabolas. The surface is computed entirely from the equation of state without empirical calibration. (D) Phase transition signature: derivative $dT_{cat}/d\rho$ versus density. The sign change from negative (free-flow, T_{cat} decreasing smoothly) to sharply negative (congestion onset) at the critical density ρ_c identifies the phase transition point. The discontinuity in the second derivative is the hallmark of a continuous (second-order) phase transition, consistent with the partition extinction theorem: at ρ_c , vehicles begin to phase-lock (convoy formation), and partition operations between adjacent vehicles undergo extinction.

10 Experimental Predictions

The vehicular equation of state makes five quantitative, testable predictions that can be verified with existing traffic data.

10.1 Prediction 1: Velocity–Partition Transition Rate

Theorem 10.1 (Velocity–Partition Transition Relationship). *The rate of categorical state transitions $\dot{N}_{\text{trans}}$ is related to the mean velocity $\bar{v}$ by*

$$\dot{N}_{\text{trans}} = \frac{\bar{v}}{L_{\text{cell}}(n)}, \quad (85)$$

where $L_{\text{cell}}(n) = L_{\text{road}}/3^n$ is the characteristic length of a partition cell at hierarchy level n .

This is measurable from OBD-II (On-Board Diagnostics) data: the velocity $\bar{v}$ is directly available, and the transition rate $\dot{N}_{\text{trans}}$ can be counted from GPS position logs by identifying changes in road hierarchy, heading, lane, and chirality. The prediction is that $\dot{N}_{\text{trans}} \propto \bar{v}$ with the proportionality constant determined by the road network partition structure.

10.2 Prediction 2: Road Capacity from Equation of State

The equation of state predicts the maximum traffic flow (road capacity) as the flow rate at which the phase transition from free-flow to synchronized flow occurs. From Theorem 5.8, the critical density is $\rho_{c1} = 1/(3V_{\text{excl}}^{\text{eff}})$. The capacity (maximum flow) is

$$q_{\text{max}} = \rho_{c1} \cdot v(\rho_{c1}) = \frac{v_{\text{free}}}{3V_{\text{excl}}^{\text{eff}}} \left(1 - \frac{1}{3}\right) = \frac{2v_{\text{free}}}{9V_{\text{excl}}^{\text{eff}}}. \quad (86)$$

For a highway lane with $v_{\text{free}} = 120$ km/h and $V_{\text{excl}}^{\text{eff}} = 50$ m (vehicle length plus safety distance at 120 km/h), this gives $q_{\text{max}} \approx 2133$ veh/h/lane, which should be compared with the Highway Capacity Manual value of ~ 2200 veh/h/lane [25].

10.3 Prediction 3: S-Entropy Signatures

Different driving scenarios produce distinct S-entropy signatures that are measurable from sensor logs:

Scenario	S_k	S_t	S_e
Highway cruising	< 0.1	< 0.1	< 0.15
Urban driving	< 0.15	$0.5\text{--}0.8$	$0.3\text{--}0.6$
Parking	$0.3\text{--}0.6$	$0.7\text{--}0.95$	$0.6\text{--}0.9$
Off-road	$0.6\text{--}0.9$	$0.3\text{--}0.6$	$0.5\text{--}0.8$

These values are computed from the definitions (Definitions 4.1–4.5) using typical sensor noise levels, velocity profiles, and energy patterns for each scenario.

10.4 Prediction 4: Phase Transition Densities

The equation of state predicts critical densities for the traffic phase transitions:

$$\rho_{c1} = \frac{1}{3V_{\text{excl}}^{\text{eff}}} \approx 22 \text{ veh/km/lane} \quad (\text{free-flow} \rightarrow \text{synchronized}), \quad (87)$$

$$\rho_{c2} = \frac{1}{V_{\text{excl}}^{\text{eff}}} \approx 67 \text{ veh/km/lane} \quad (\text{synchronized} \rightarrow \text{jam}). \quad (88)$$

These should be compared with empirical critical densities: $\rho_{c1}^{\text{emp}} \approx 20$ veh/km/lane and $\rho_{c2}^{\text{emp}} \approx 70$ veh/km/lane from highway detector data [5, 4].

10.5 Prediction 5: Partition Depth of Road Networks

The ternary tree structure of the road hierarchy predicts that the partition depth of a road network with N_{segments} road segments is

$$M_{\text{road}} = \log_3 N_{\text{segments}}. \quad (89)$$

For Munich ($N_{\text{segments}} \approx 50000$), this gives $M_{\text{road}} \approx 9.9$, predicting approximately 10 hierarchy levels. This can be compared with the number of levels in the OpenStreetMap road classification system (motorway, trunk, primary, secondary, tertiary, residential, service, track, path, footway—approximately 10 levels), and with hierarchical road databases used in navigation systems.

11 Conclusion

We have derived, from the single axiom that the vehicular phase space is bounded ($V_{\Gamma} < \infty$), the complete equations of state governing a vehicle’s trajectory. The logical chain is:

1. Bounded phase space (Axiom 2.1) $\implies$
2. Finite state count $N_{\text{max}} = V_{\Gamma}/h^3$ (Theorem 2.6) $\implies$
3. Poincaré recurrence (Theorem 2.8) $\implies$
4. Oscillatory dynamics with categorical description via partition coordinates (n, ℓ, m, s) with $C(n) = 2n^2$ (Theorem 3.5) $\implies$
5. S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ providing complete description (Theorem 4.7) $\implies$
6. Vehicular equation of state $P_{\text{drive}} V_{\text{road}} = N k_B T_{\text{cat}} S$ (Theorem 5.6, Theorem 5.10) $\implies$
7. Zero categorical Lyapunov exponent (Corollary 6.3) $\implies$

8. Trajectory completion in $O(\log_3 N)$ (Theorem 8.3).

The equation of state reproduces classical traffic flow theory (Greenshields, Lighthill–Whitham–Richards) as limiting cases and predicts congestion as a genuine phase transition with quantitative critical densities consistent with empirical data. The S-entropy evolution equations, the Position-Partition Bijection, and the completion morphism structure provide the complete mathematical apparatus for trajectory determination without forward simulation.

The key conceptual advance is the replacement of *prediction* by *completion*. A vehicle’s trajectory is not predicted by integrating differential equations forward in time—a process that is computationally expensive ($O(N)$) and physically unstable (positive Lyapunov exponents in physical coordinates). Instead, the trajectory is *completed* by determining the penultimate partition state and applying the unique completion morphism—a process that is computationally efficient ($O(\log_3 N)$) and categorically stable (zero Lyapunov exponent in S-entropy coordinates).

The implications for autonomous driving are immediate. Current approaches—whether rule-based [2], learning-based [28, 29], or planning-based [3]—all attempt to predict the future trajectory and then execute it. Our framework shows that prediction is both unnecessary and impossible beyond the Lyapunov horizon. What is needed instead is the equation of state: the algebraic relationship between driving pressure, road volume, and categorical temperature that determines the trajectory completion.

The five experimental predictions (velocity–transition rate, road capacity, S-entropy signatures, phase transition densities, and road network partition depth) are all testable with existing traffic data and sensor logs. Verification of these predictions would constitute strong evidence for the vehicular equation of state and, more broadly, for the bounded phase space framework as a foundation for autonomous systems.

This paper is the second in a three-paper series. The first paper [6] established the categorical framework; the present paper has derived the hard physics (equations of state). The third paper will address practical implementation: the construction of a vehicular trajectory completion system that operates in real time using the equation of state derived here, implementing the $O(\log_3 N)$ completion algorithm on the categorical operating system Buhera [15].

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